

BFS Traversal

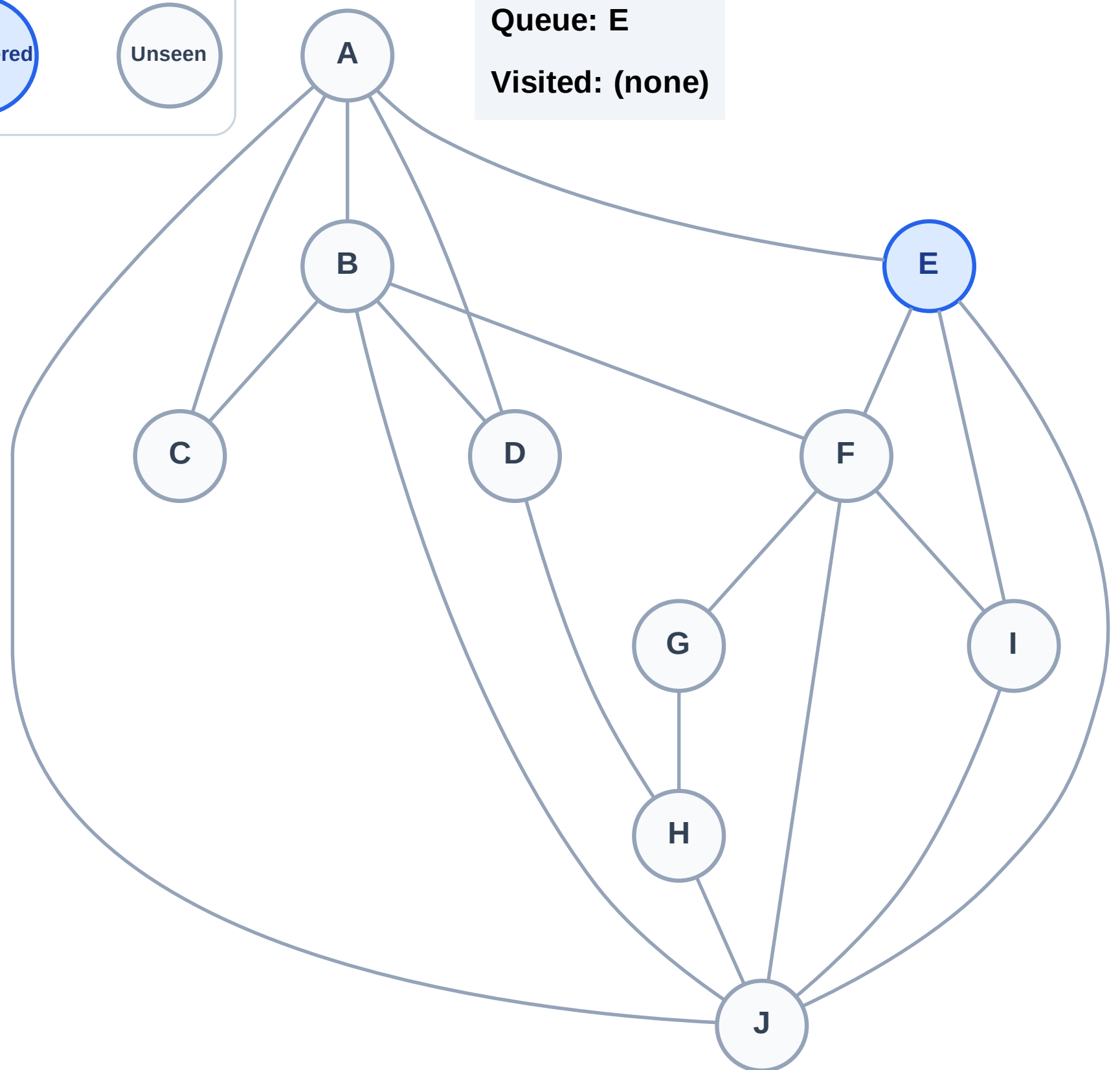
Step 1: Start at E

Legend



Queue: E

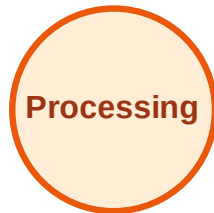
Visited: (none)



BFS Traversal

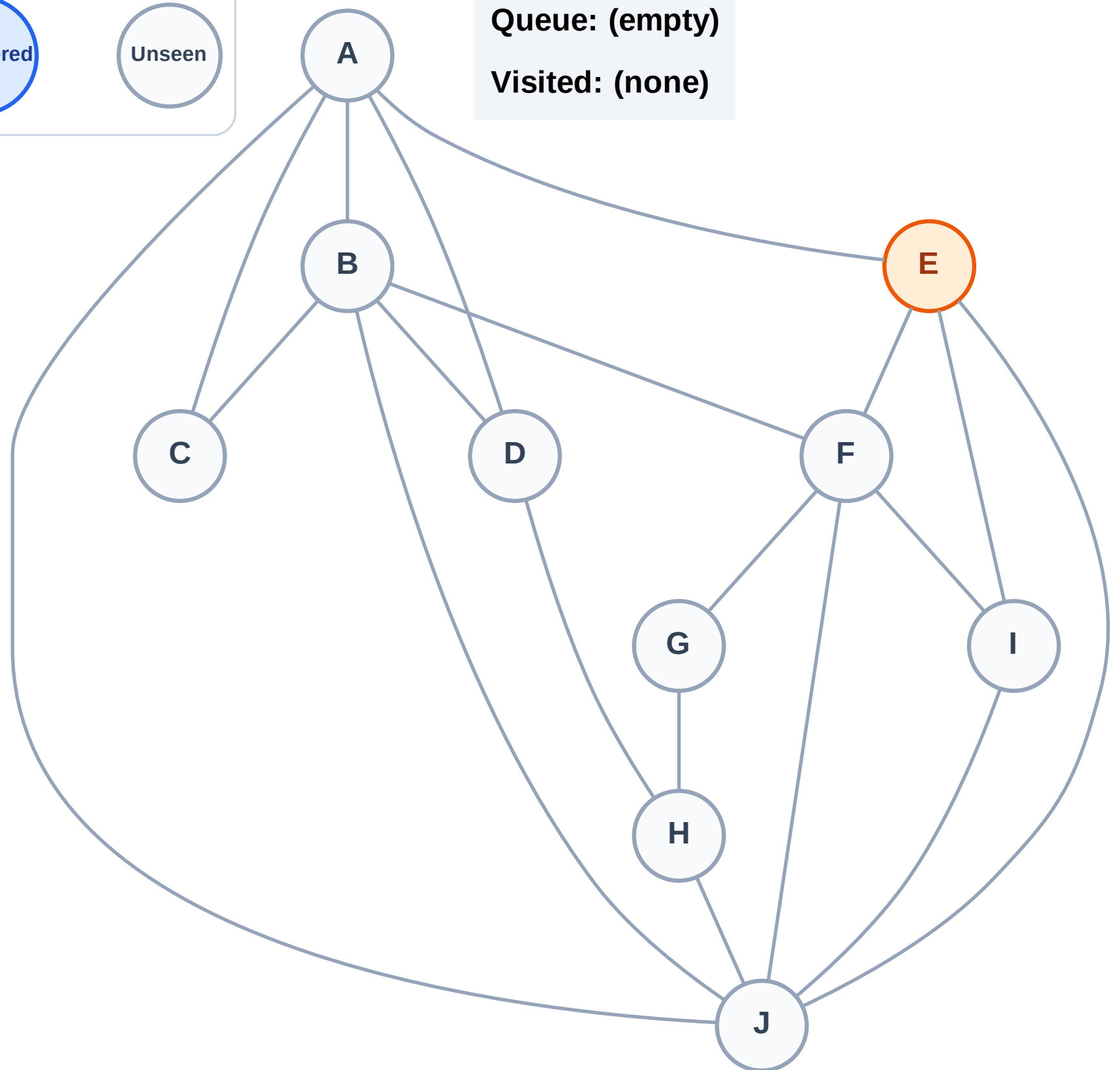
Step 2: Process node E

Legend



Queue: (empty)

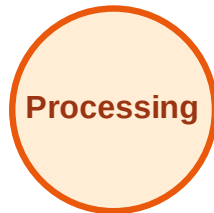
Visited: (none)



BFS Traversal

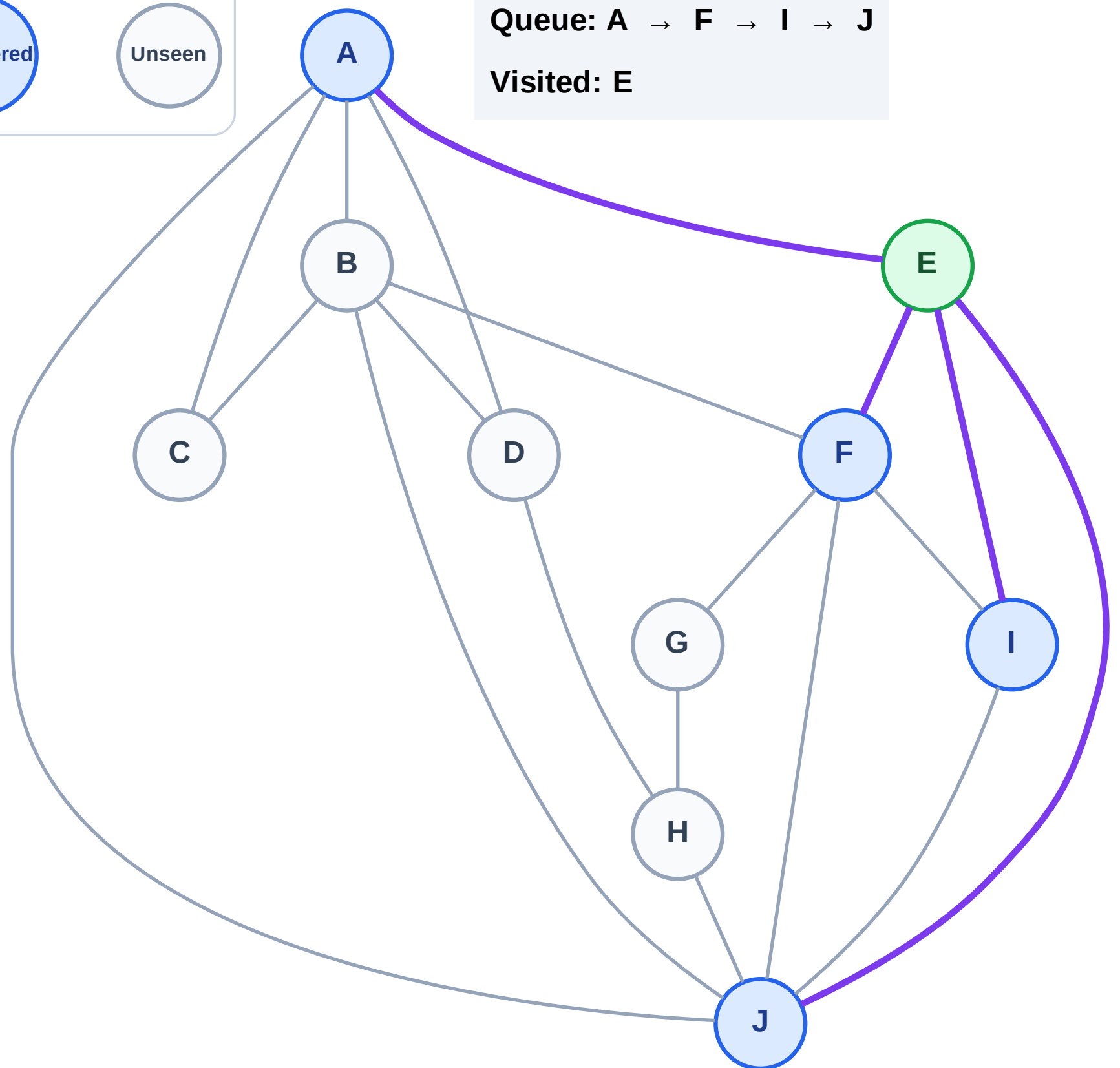
Step 3: Discover A, F, I, J from E

Legend



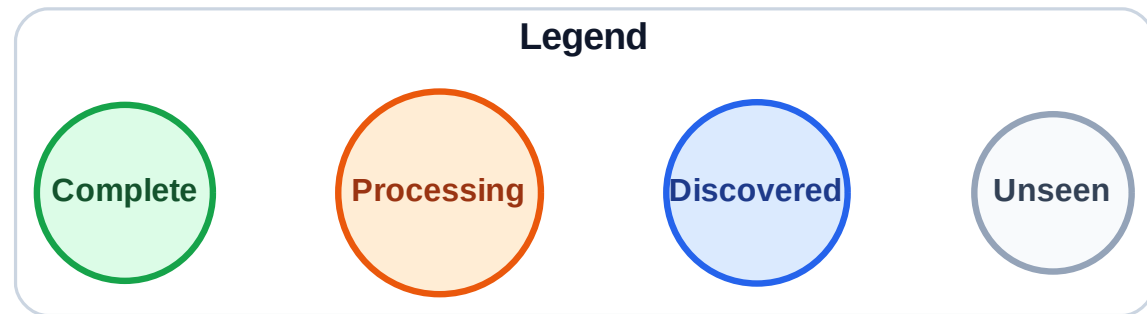
Queue: A → F → I → J

Visited: E

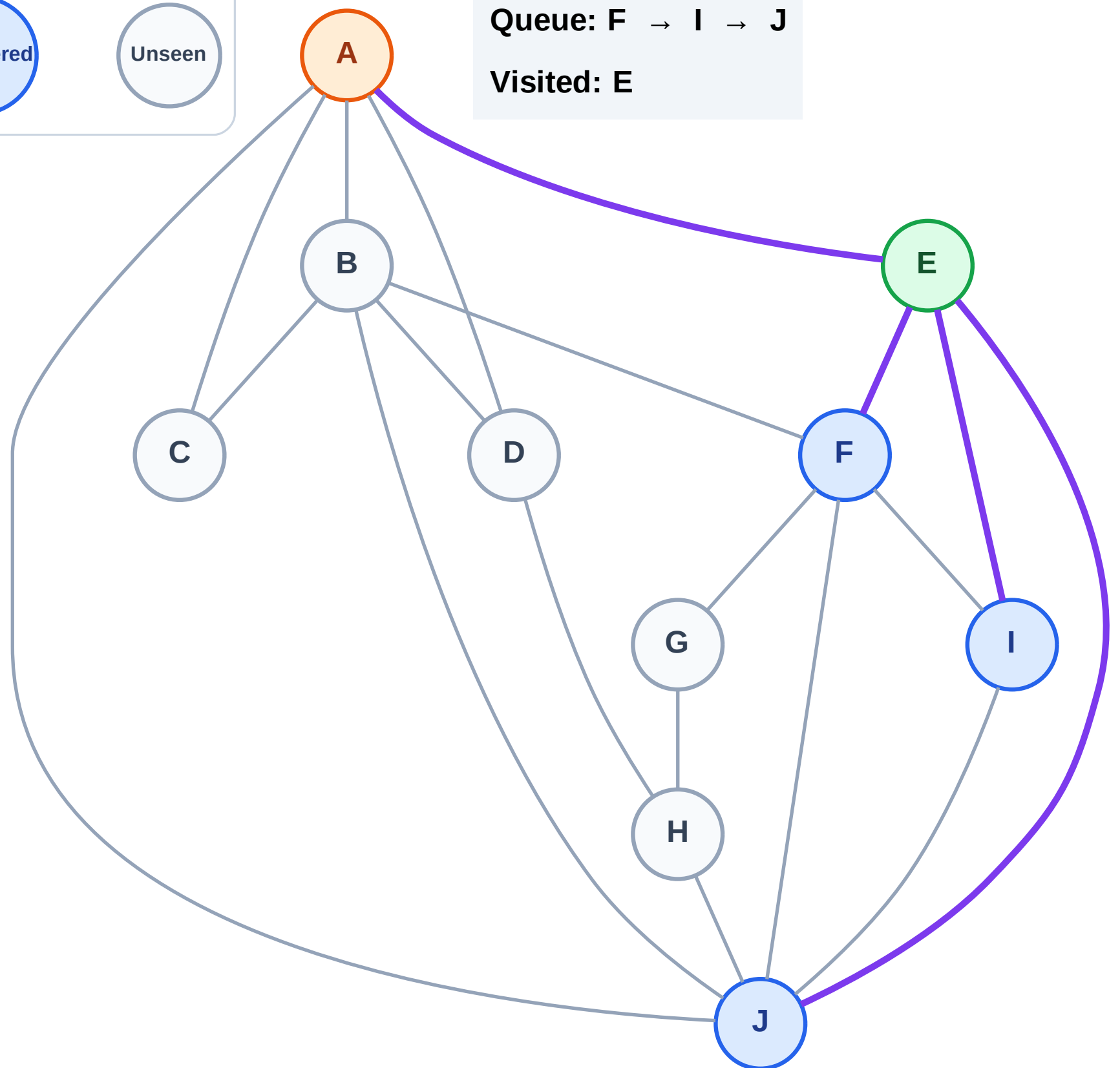


BFS Traversal

Step 4: Process node A



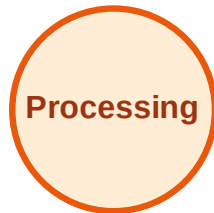
Queue: F → I → J
Visited: E



BFS Traversal

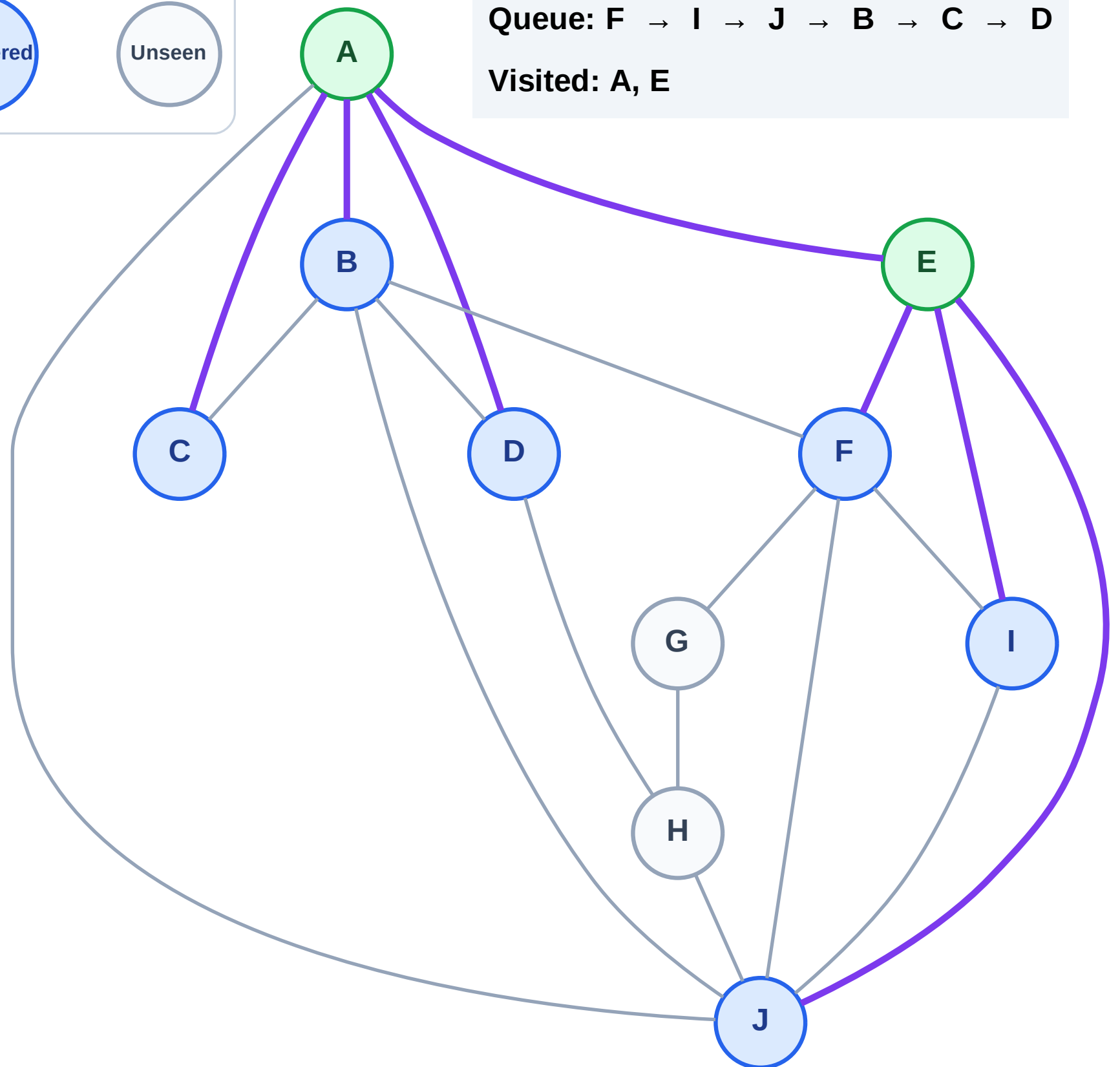
Step 5: Discover B, C, D from A

Legend



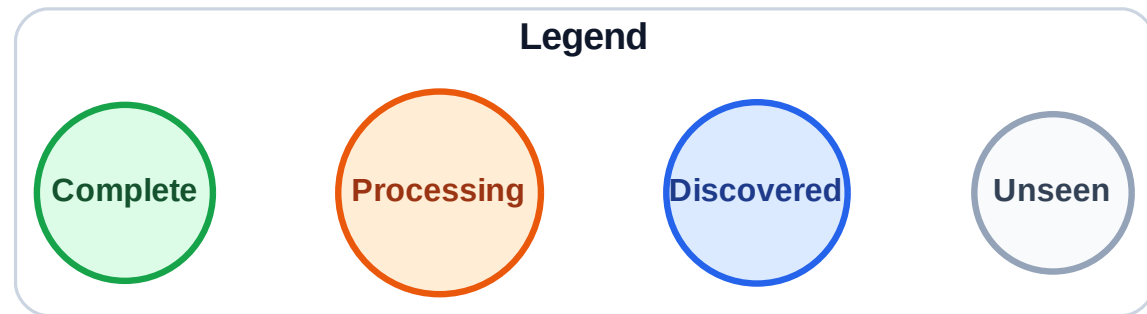
Queue: F → I → J → B → C → D

Visited: A, E



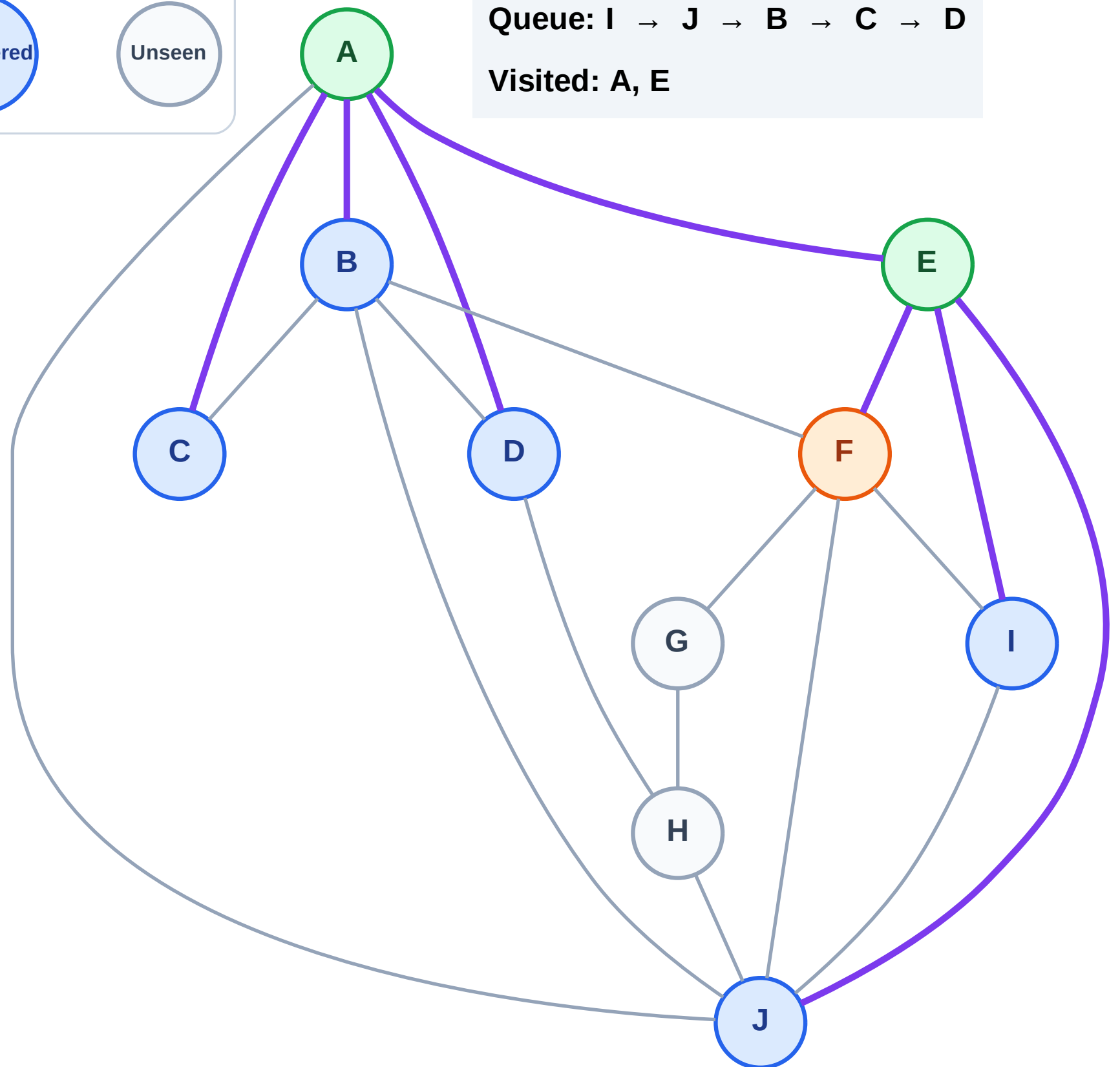
BFS Traversal

Step 6: Process node F



Queue: I → J → B → C → D

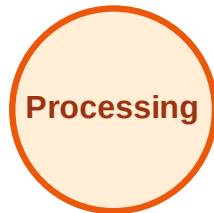
Visited: A, E



BFS Traversal

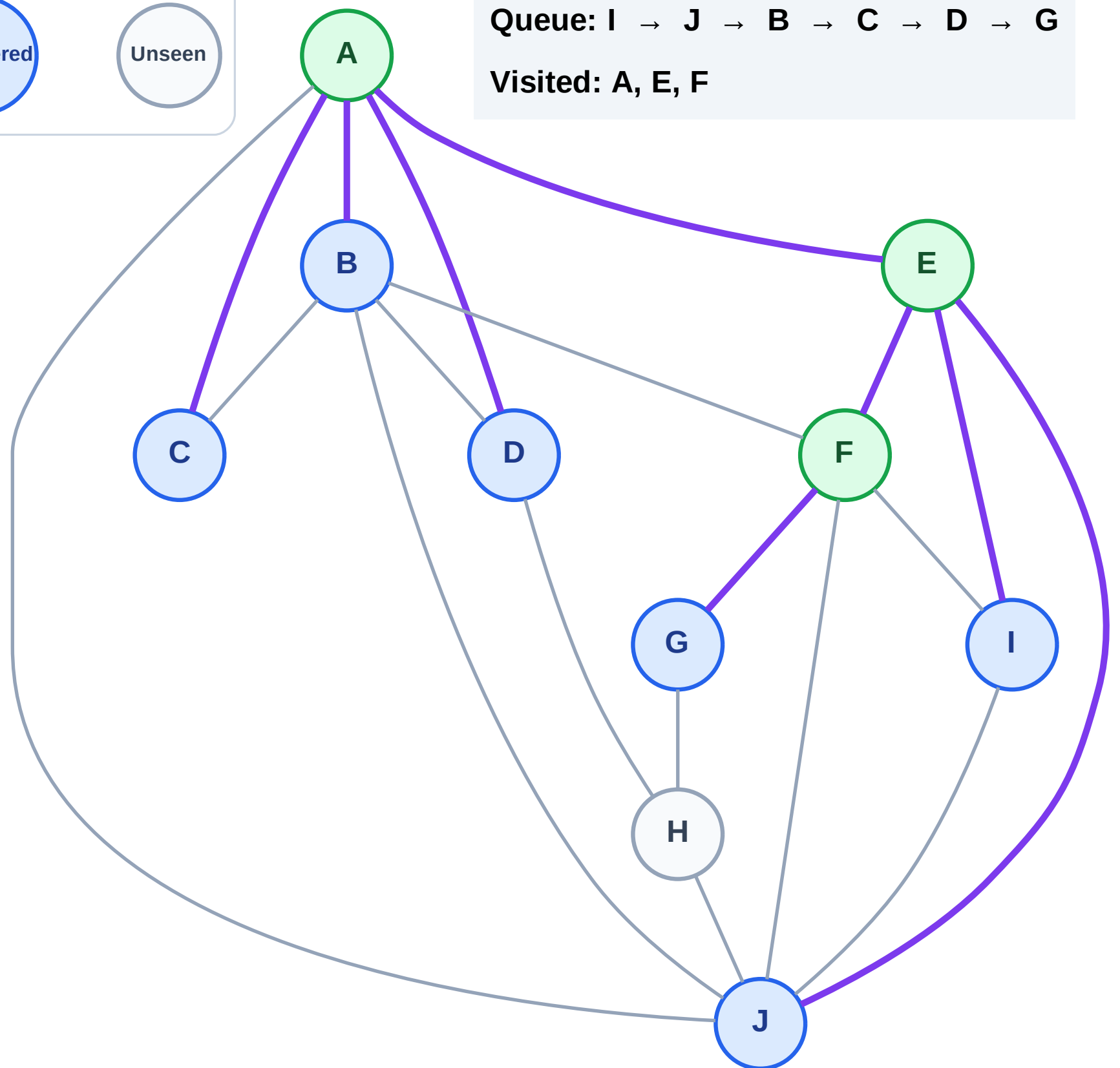
Step 7: Discover G from F

Legend



Queue: I → J → B → C → D → G

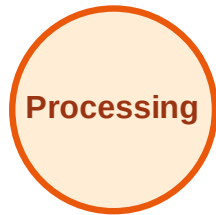
Visited: A, E, F



BFS Traversal

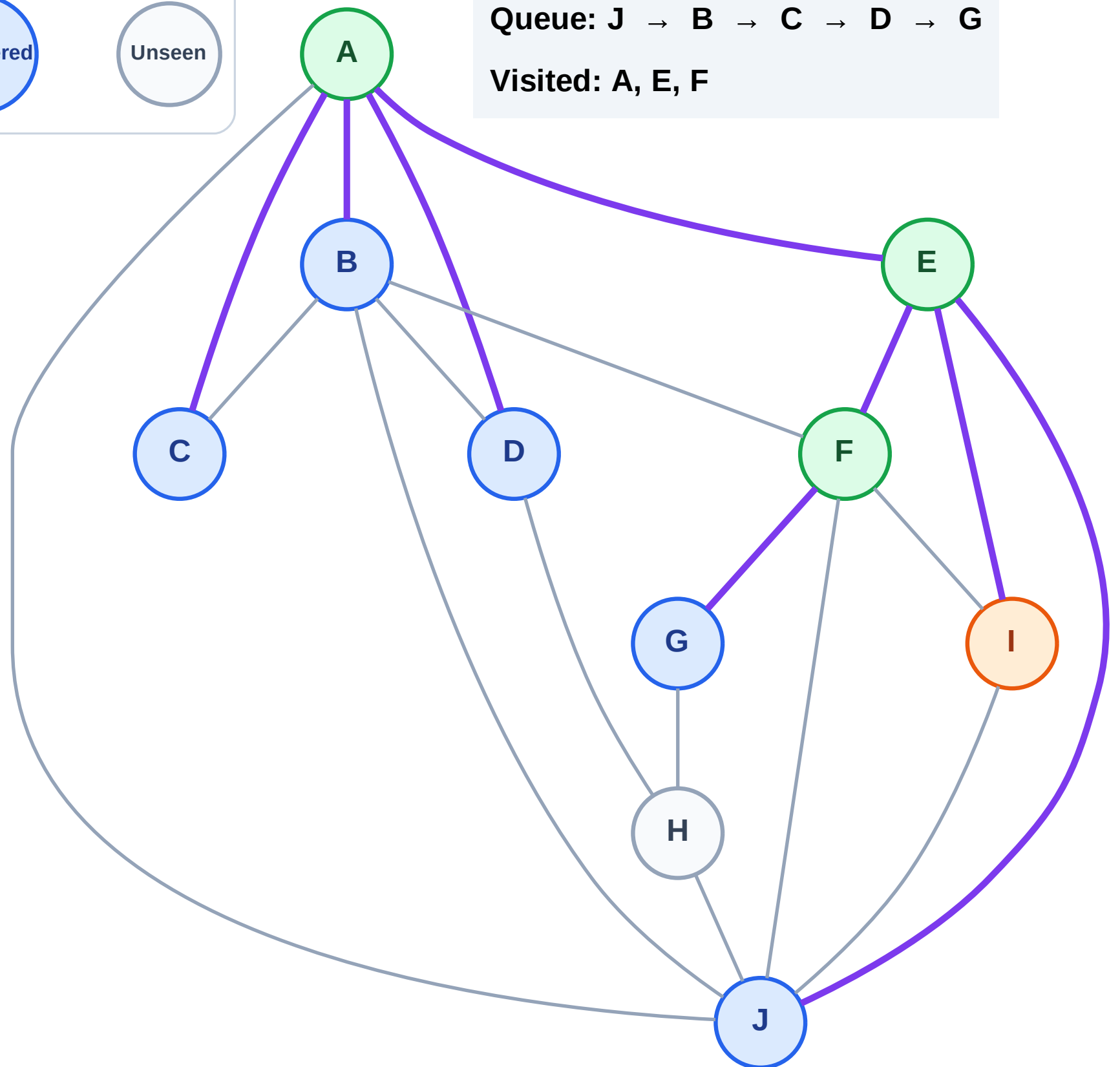
Step 8: Process node I

Legend



Queue: J → B → C → D → G

Visited: A, E, F



BFS Traversal

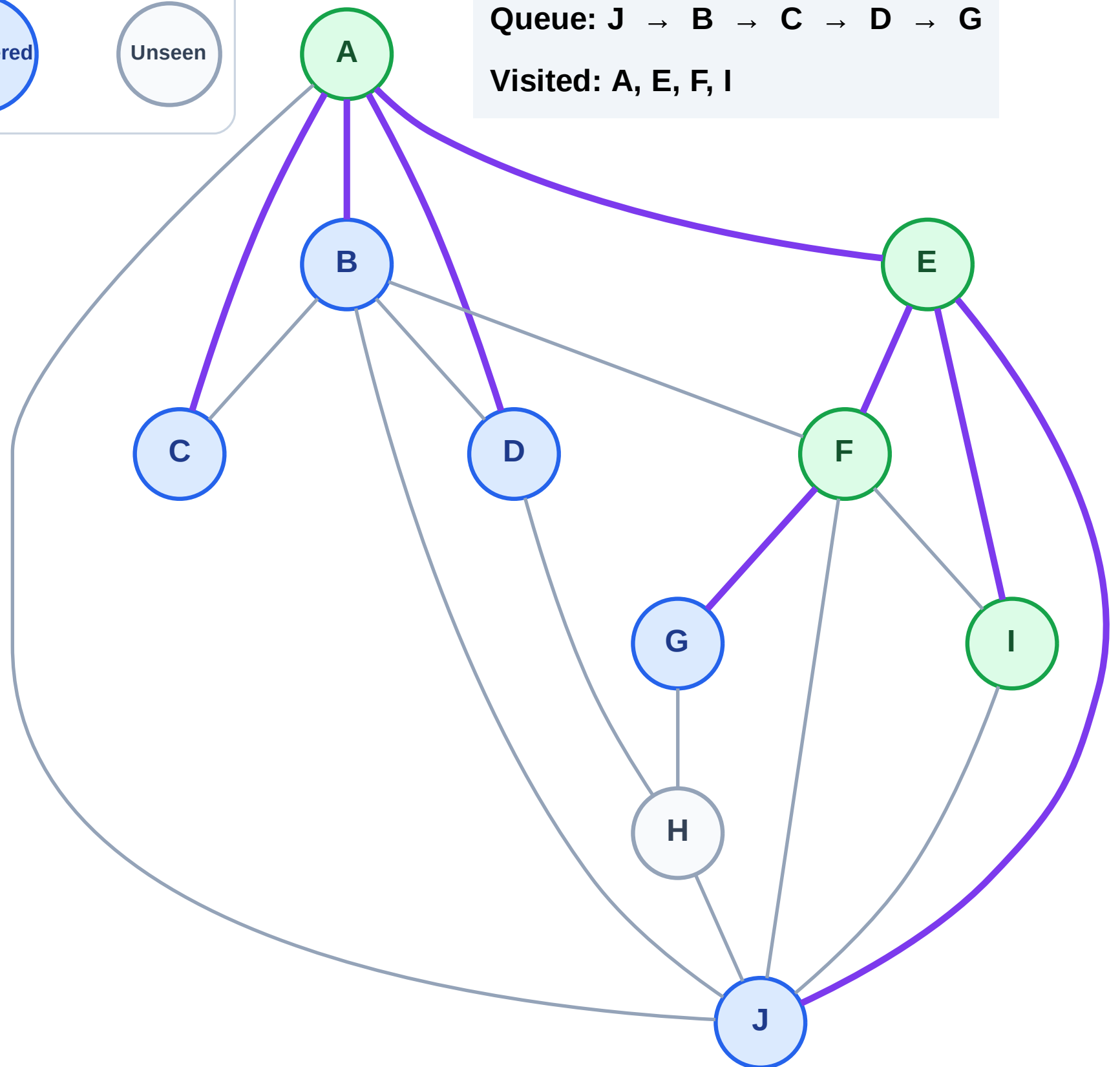
Step 9: No new neighbors from I

Legend



Queue: J → B → C → D → G

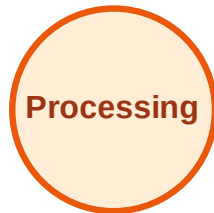
Visited: A, E, F, I



BFS Traversal

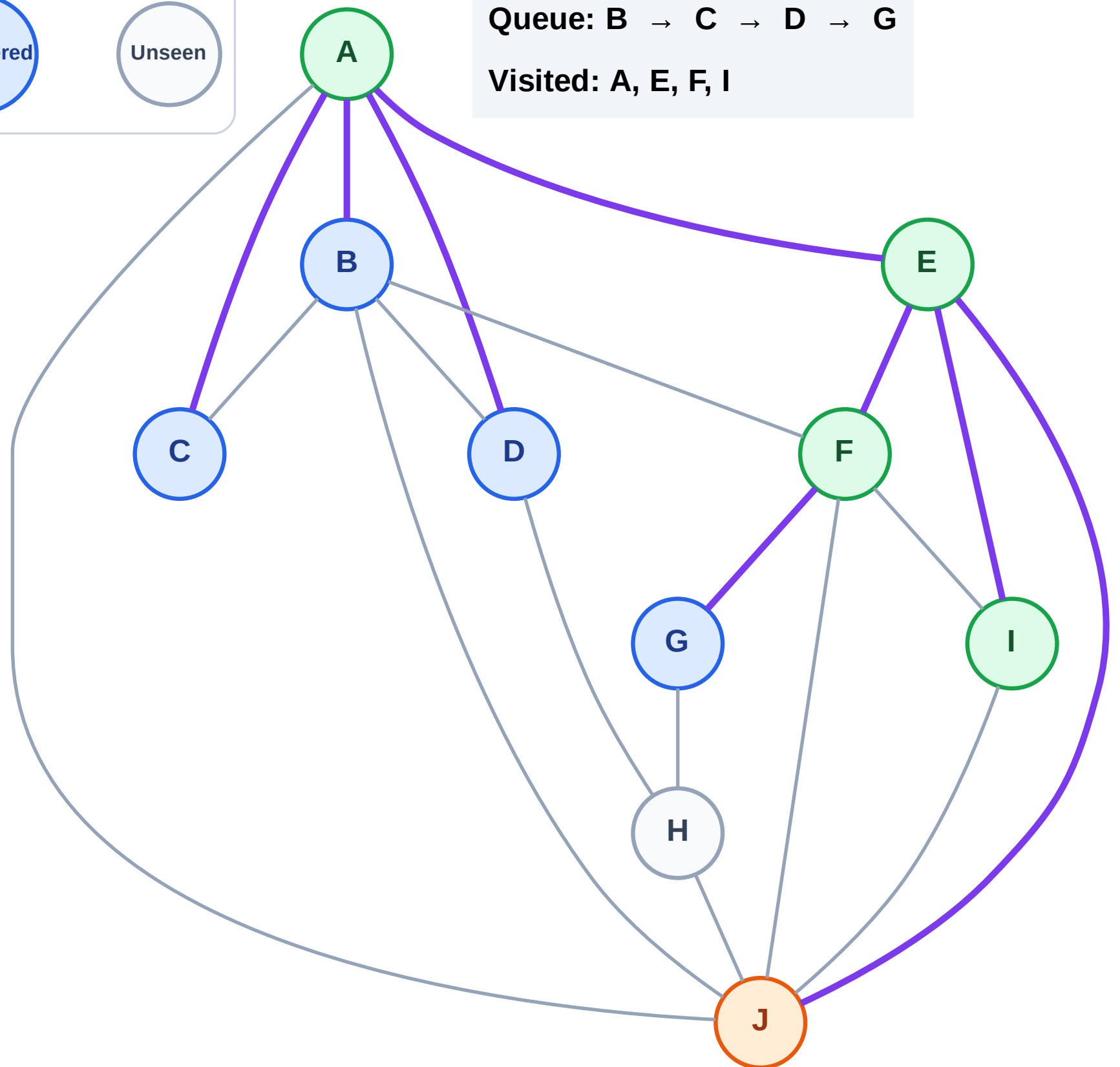
Step 10: Process node J

Legend



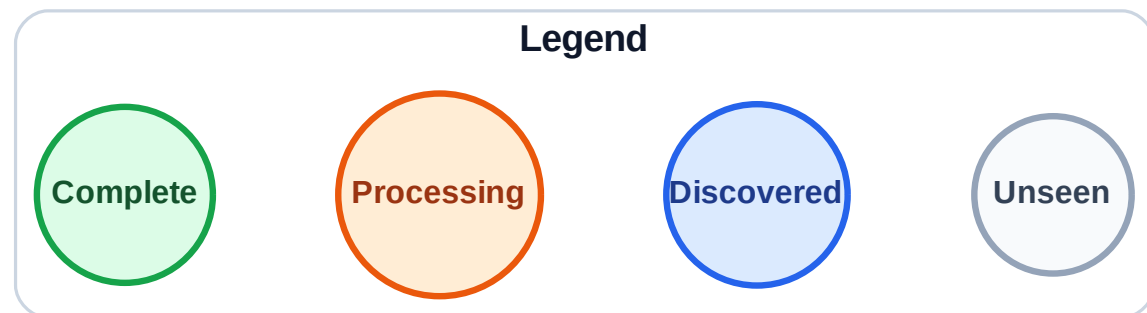
Queue: B → C → D → G

Visited: A, E, F, I



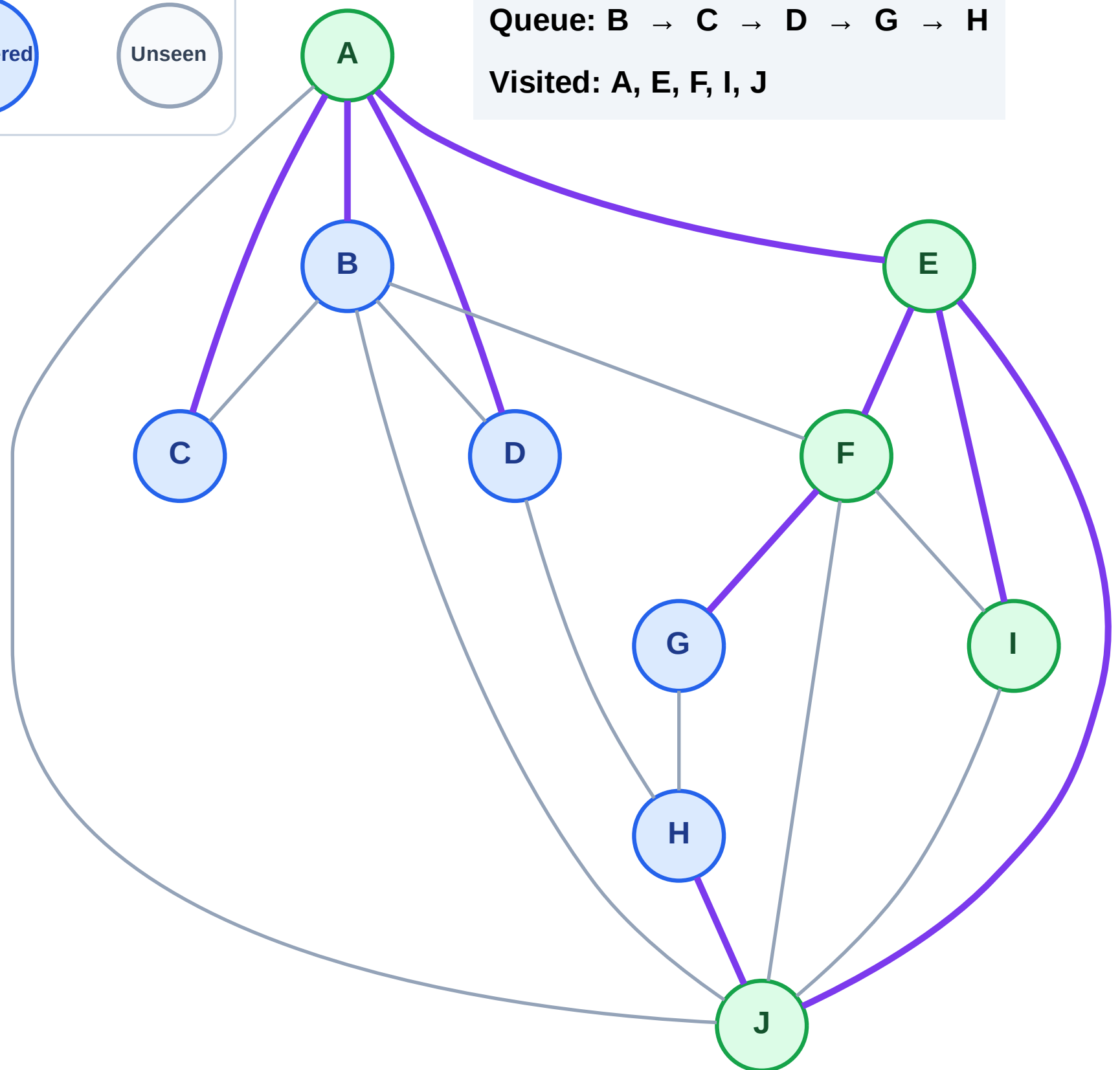
BFS Traversal

Step 11: Discover H from J



Queue: B → C → D → G → H

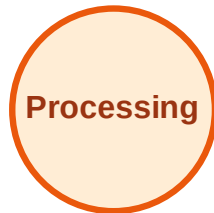
Visited: A, E, F, I, J



BFS Traversal

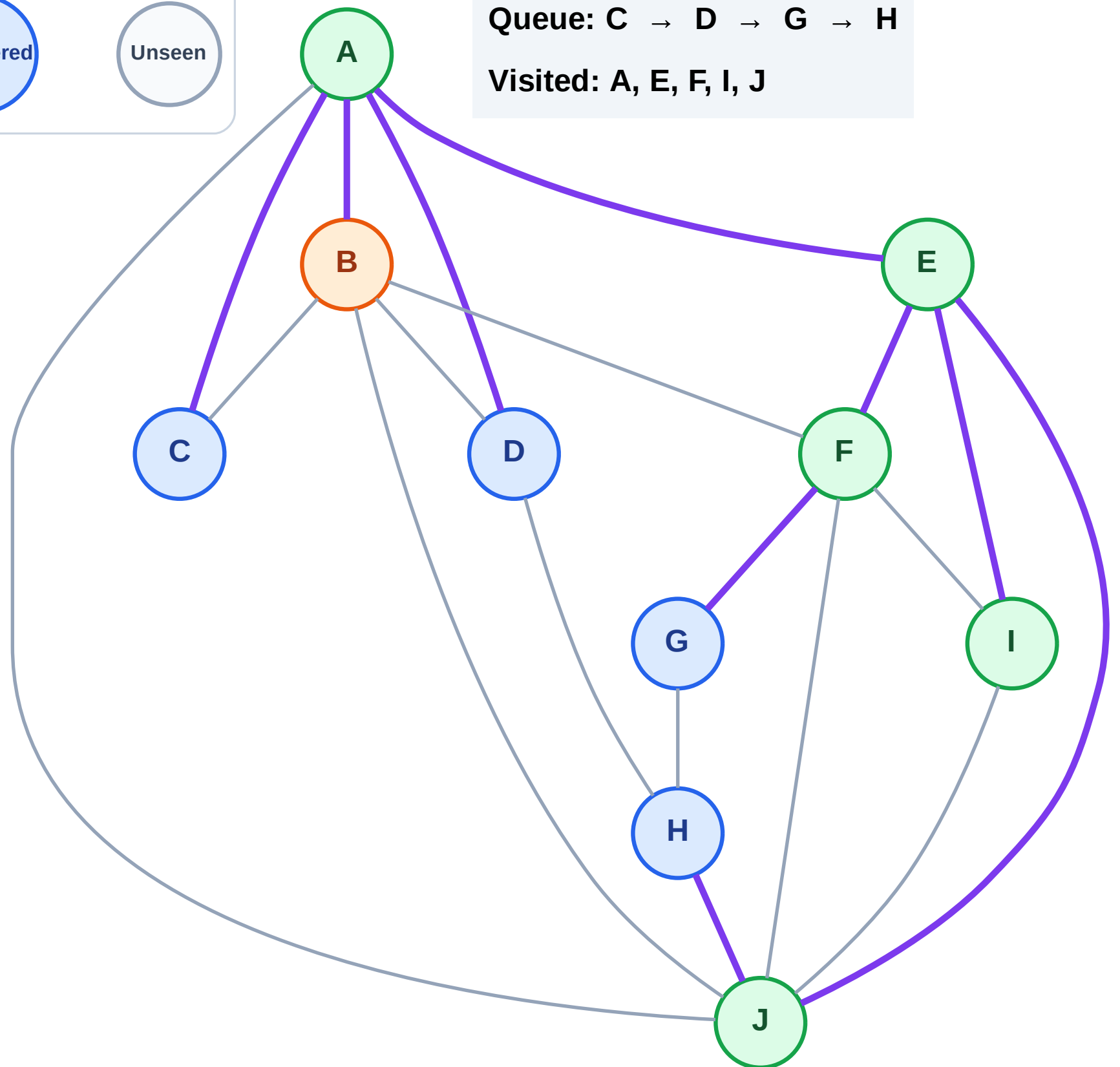
Step 12: Process node B

Legend



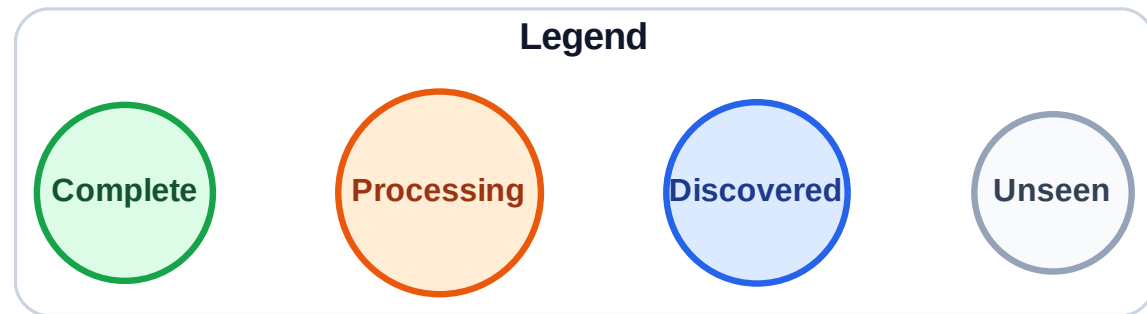
Queue: C → D → G → H

Visited: A, E, F, I, J



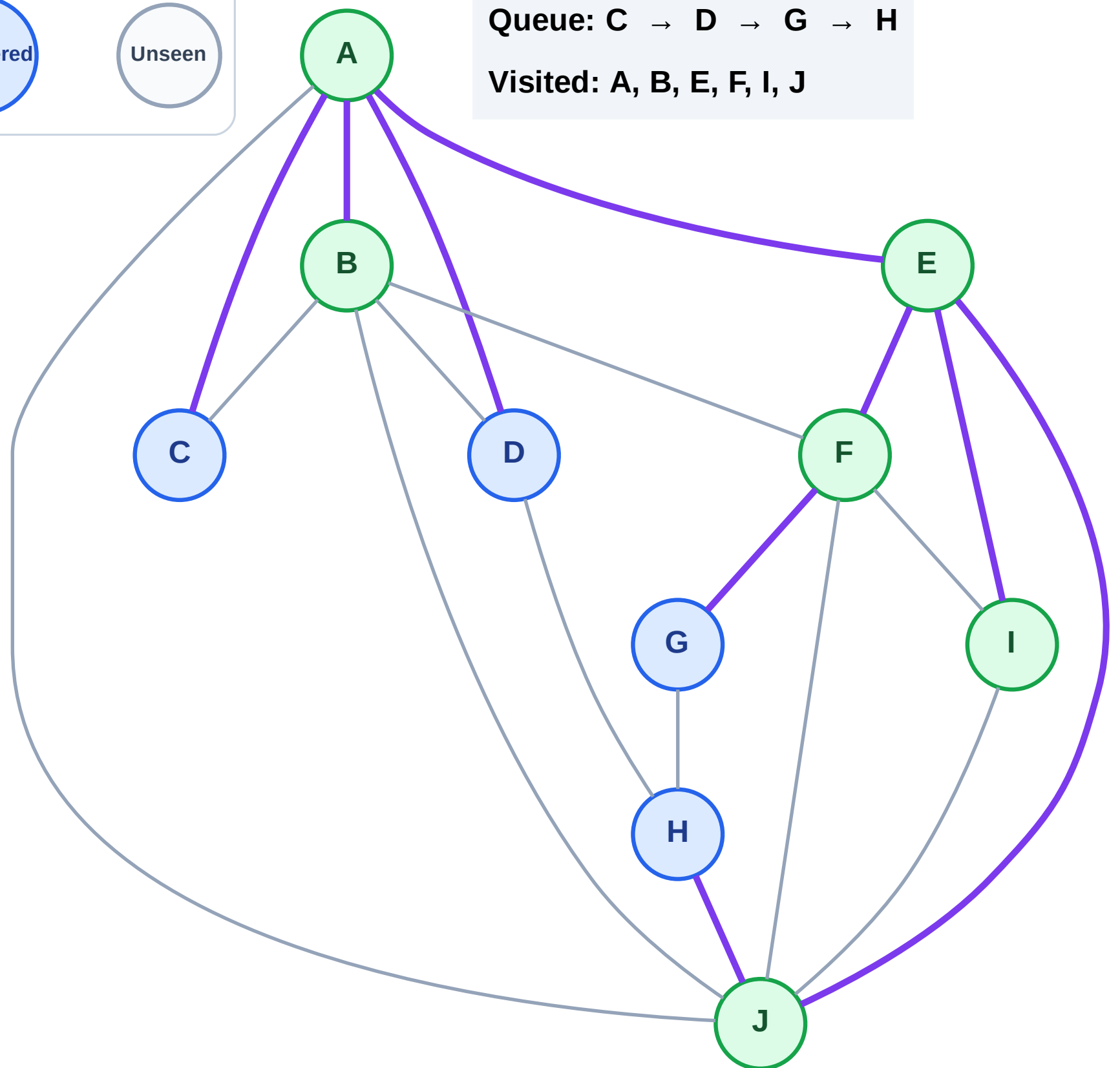
BFS Traversal

Step 13: No new neighbors from B



Queue: C → D → G → H

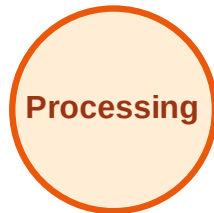
Visited: A, B, E, F, I, J



BFS Traversal

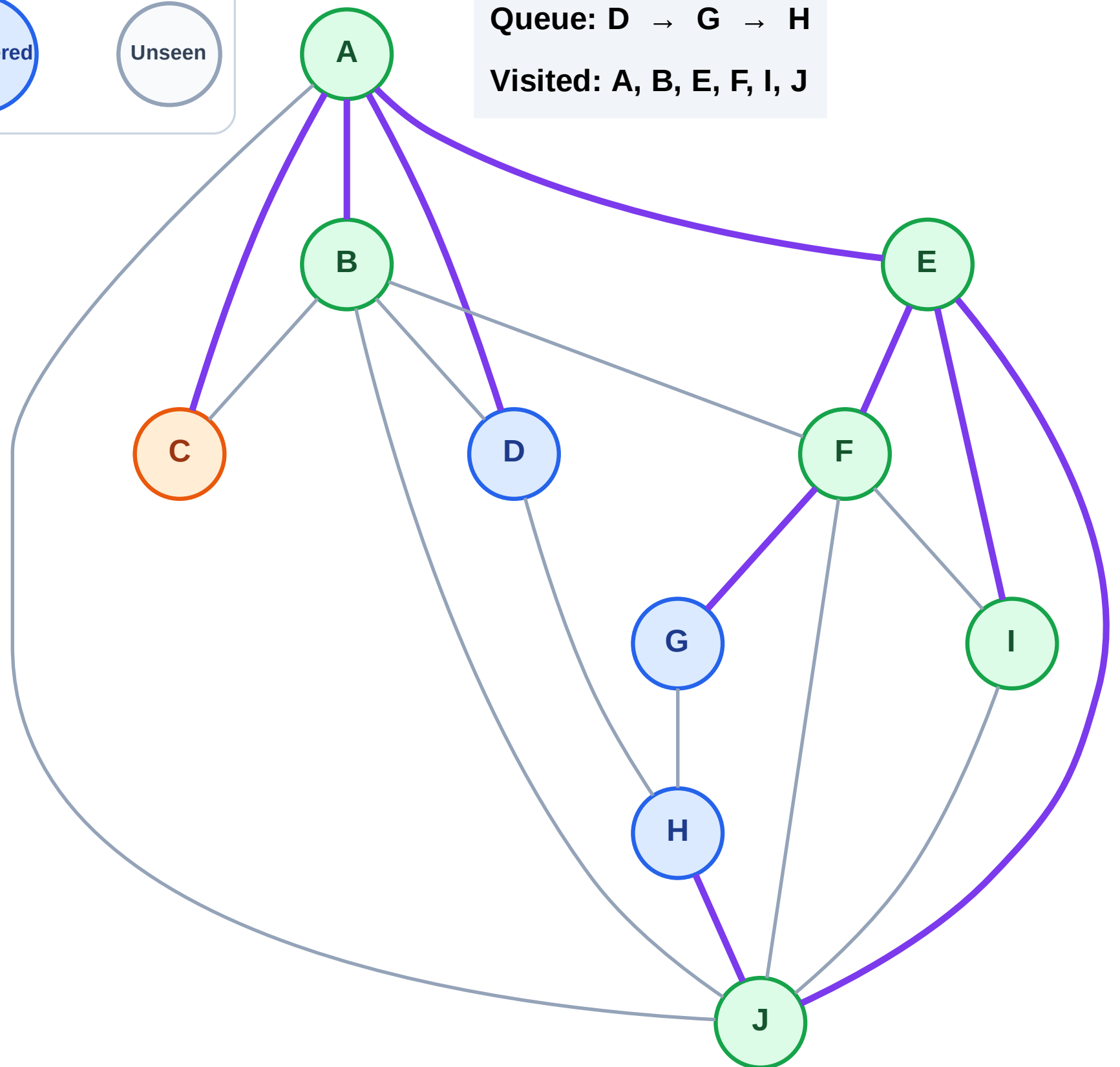
Step 14: Process node C

Legend



Queue: D → G → H

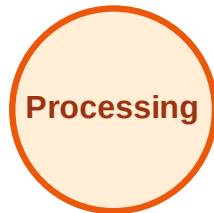
Visited: A, B, E, F, I, J



BFS Traversal

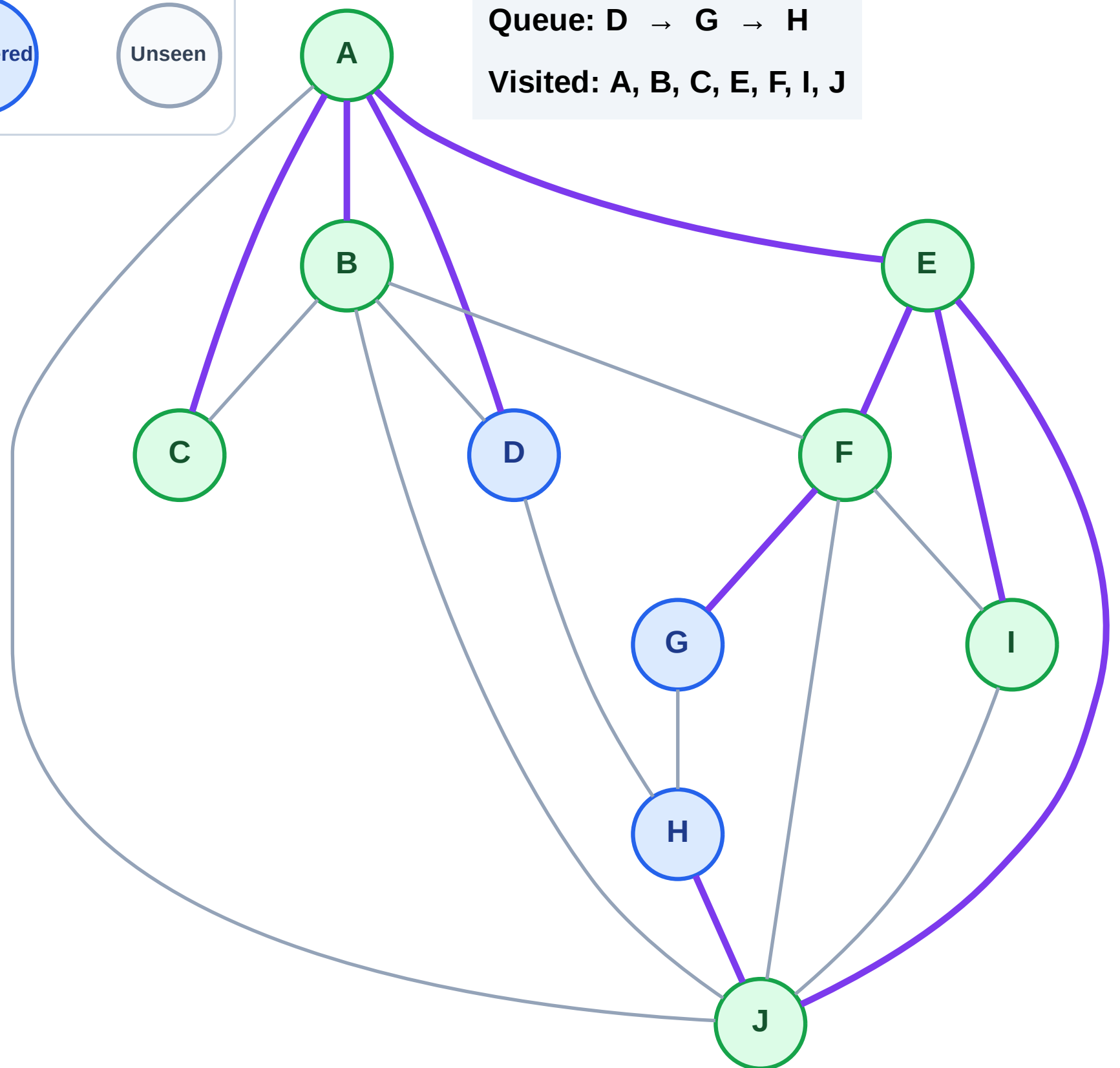
Step 15: No new neighbors from C

Legend



Queue: D → G → H

Visited: A, B, C, E, F, I, J



BFS Traversal

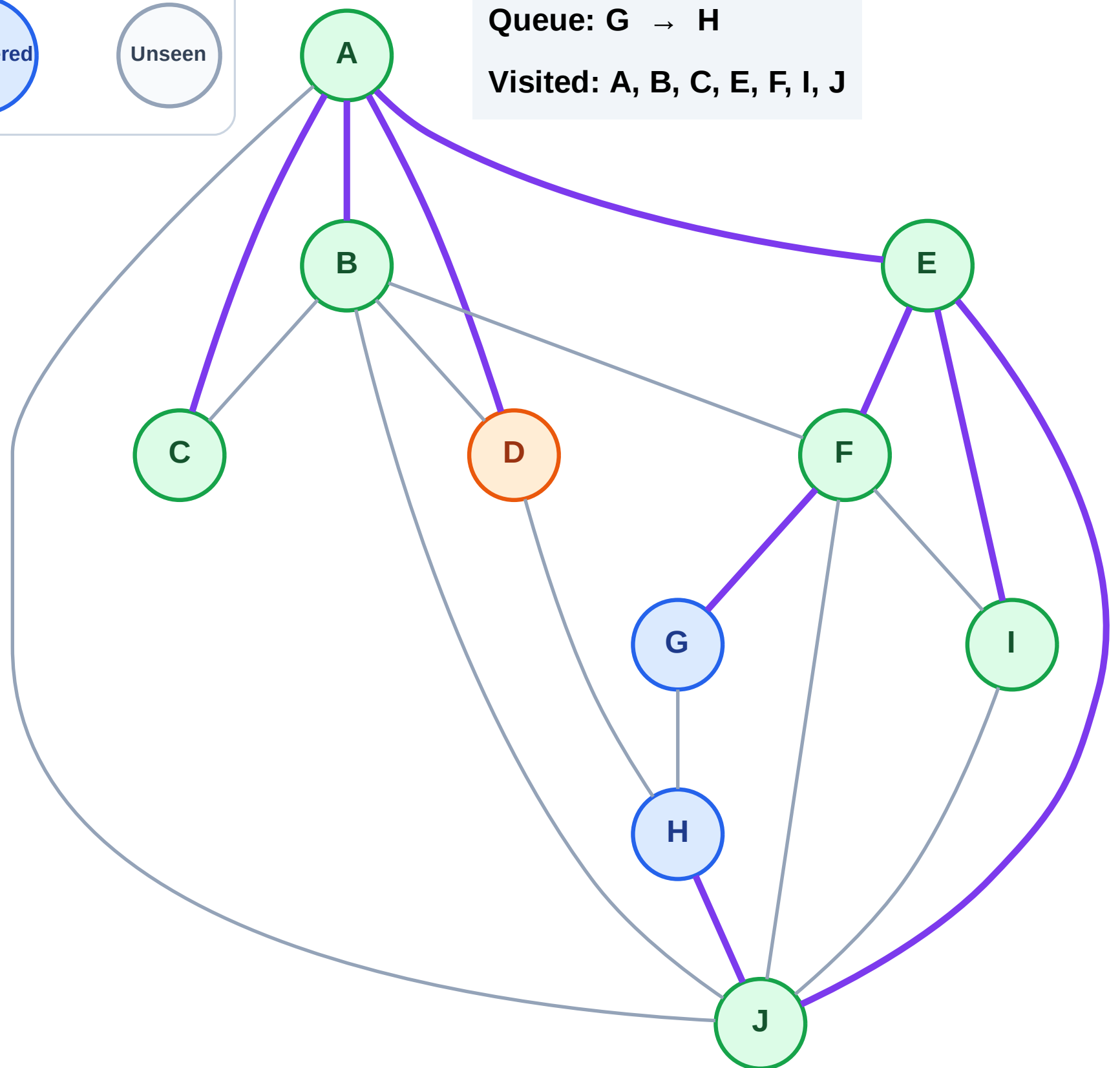
Step 16: Process node D

Legend



Queue: G → H

Visited: A, B, C, E, F, I, J



BFS Traversal

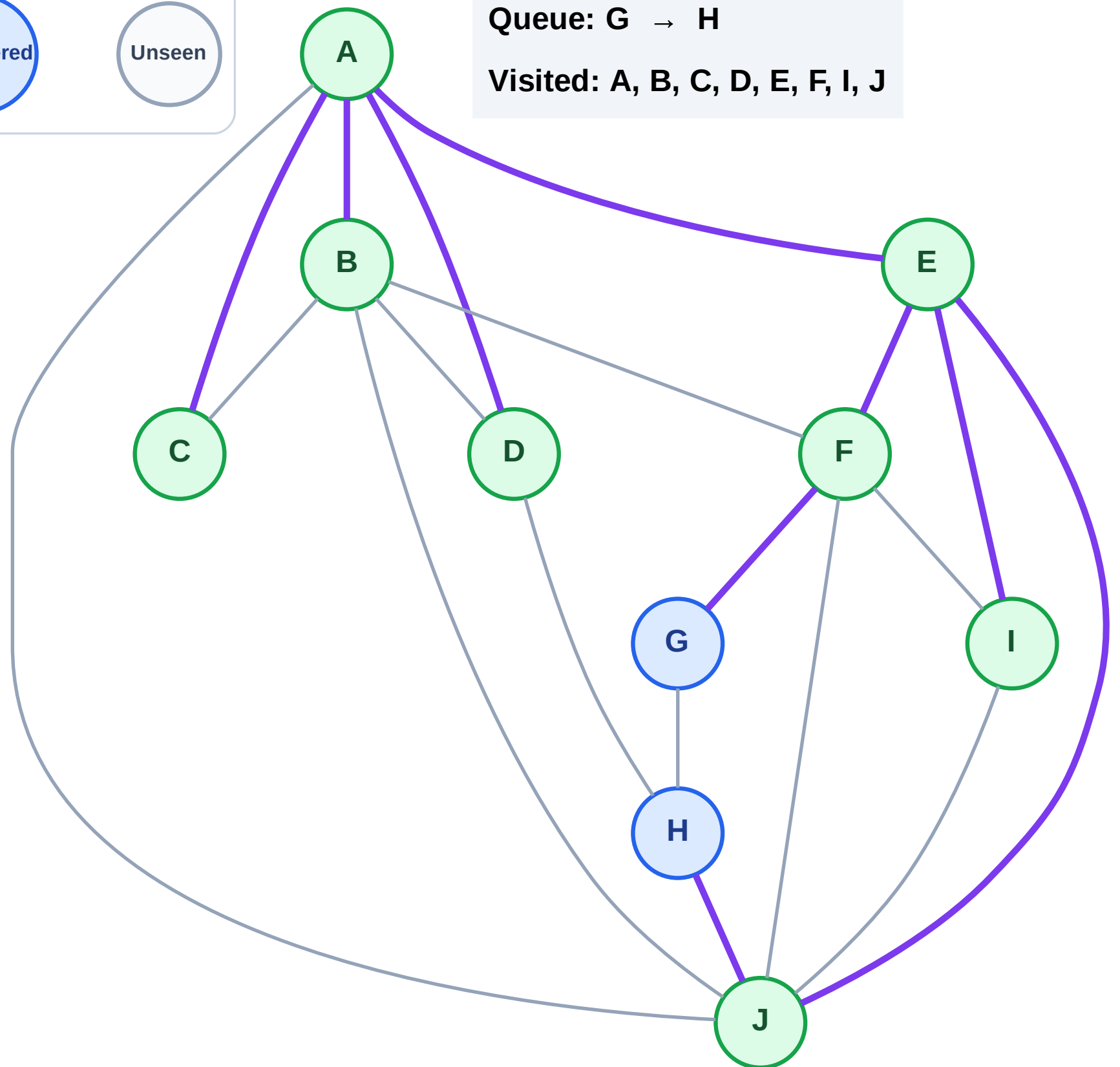
Step 17: No new neighbors from D

Legend



Queue: G → H

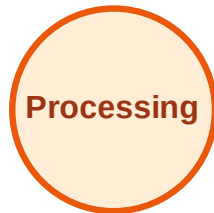
Visited: A, B, C, D, E, F, I, J



BFS Traversal

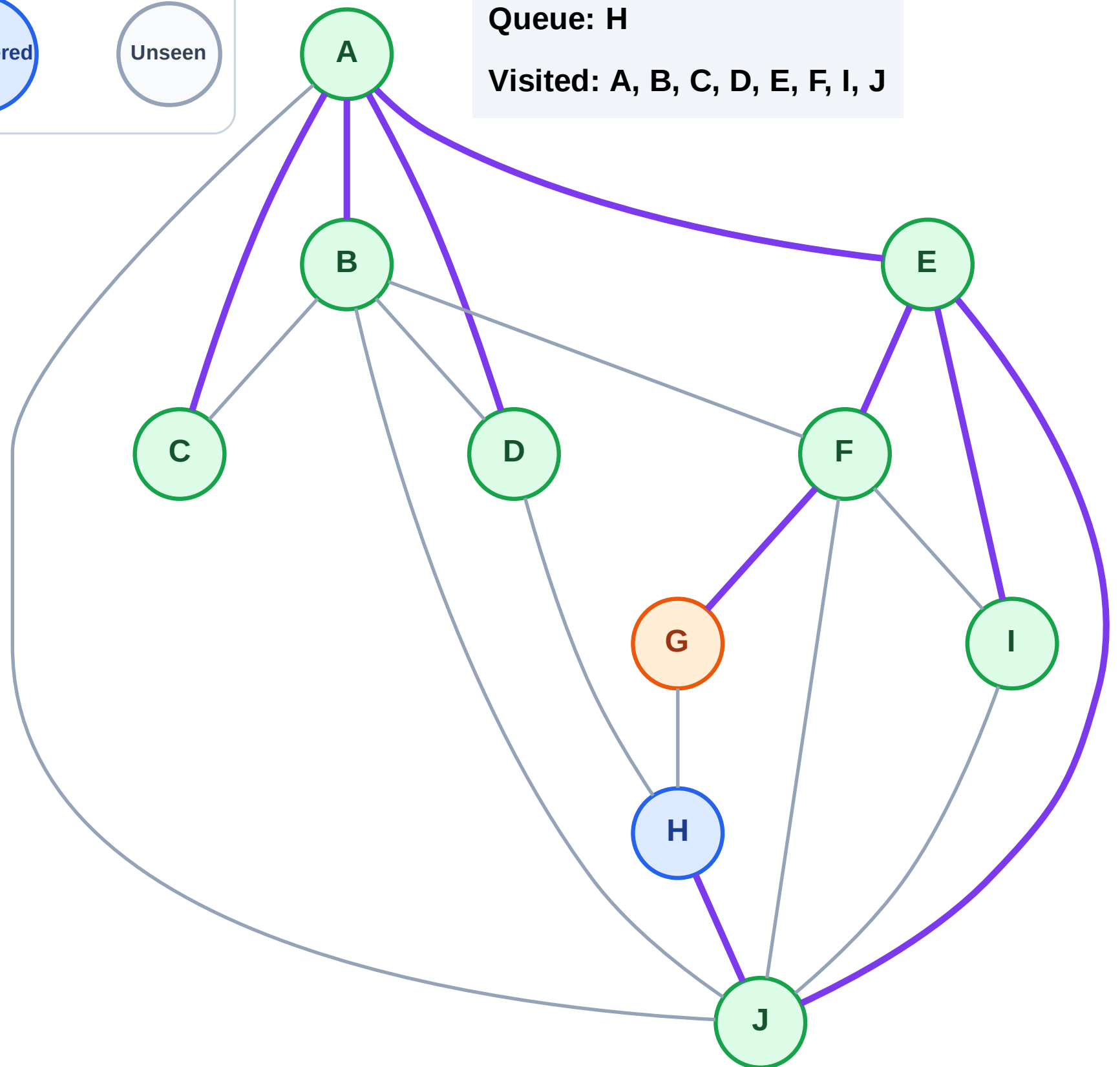
Step 18: Process node G

Legend



Queue: H

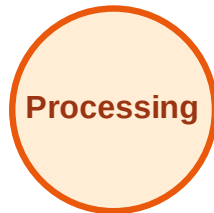
Visited: A, B, C, D, E, F, I, J



BFS Traversal

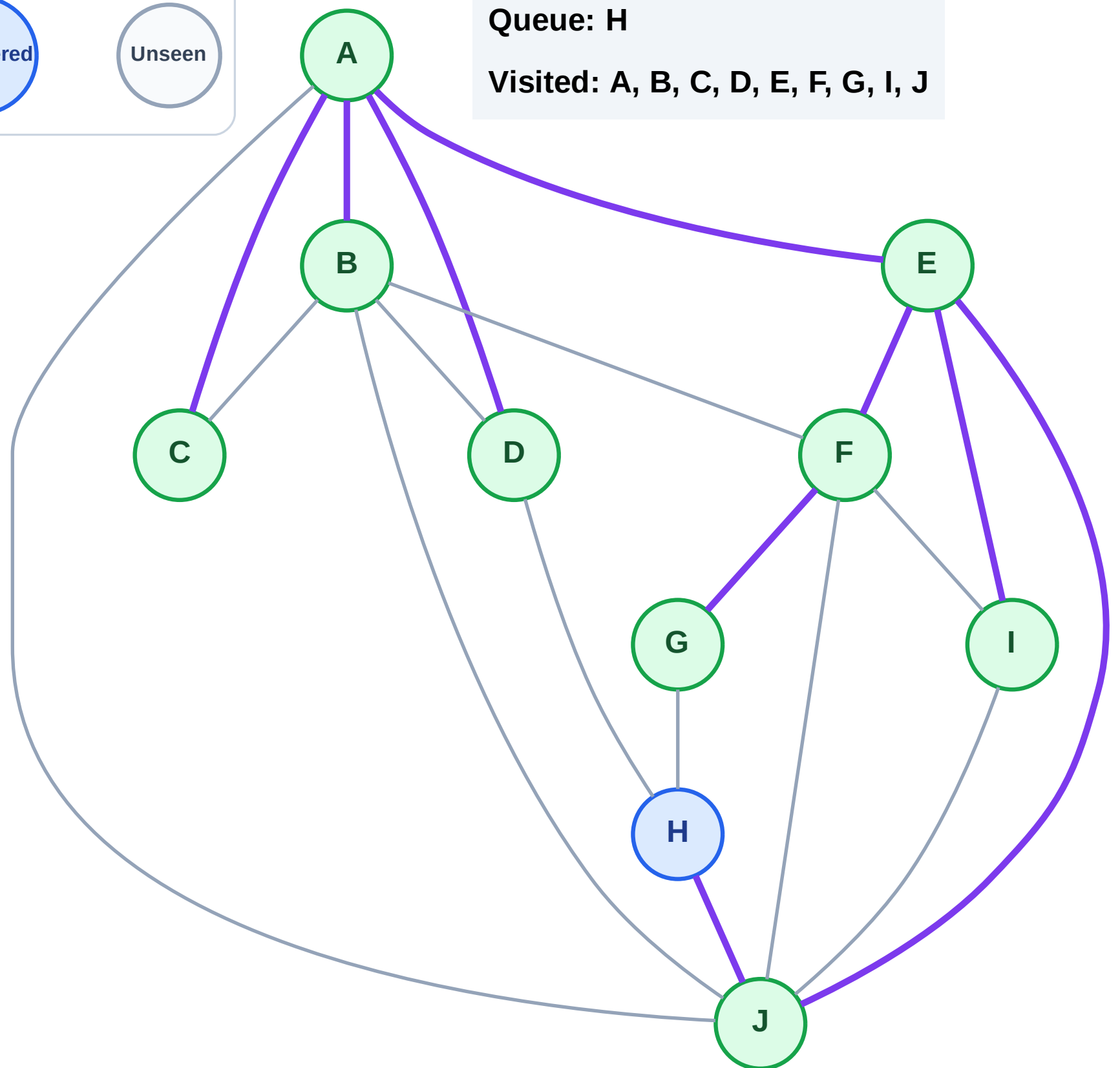
Step 19: No new neighbors from G

Legend



Queue: H

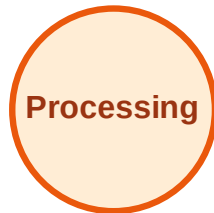
Visited: A, B, C, D, E, F, G, I, J



BFS Traversal

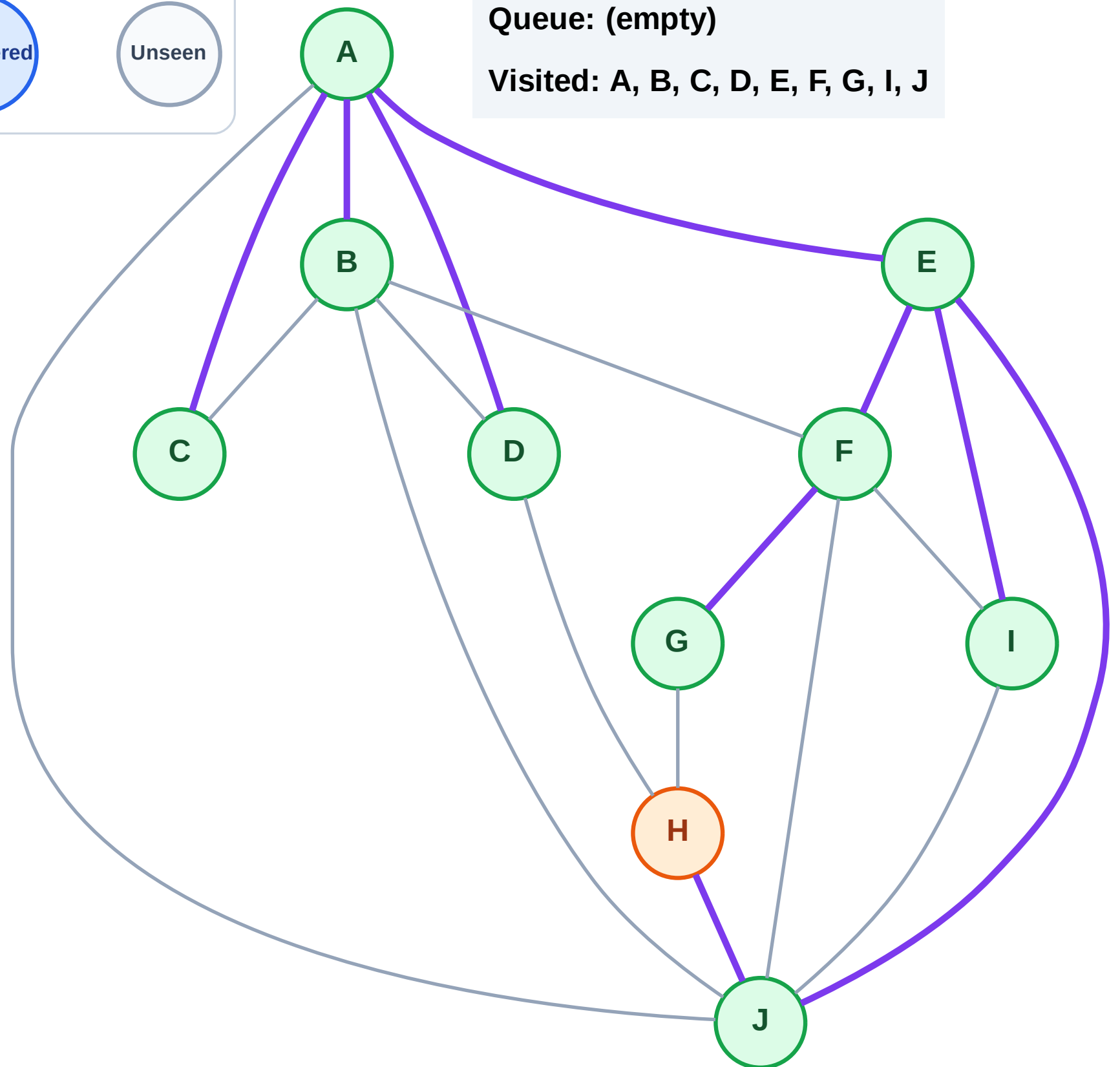
Step 20: Process node H

Legend



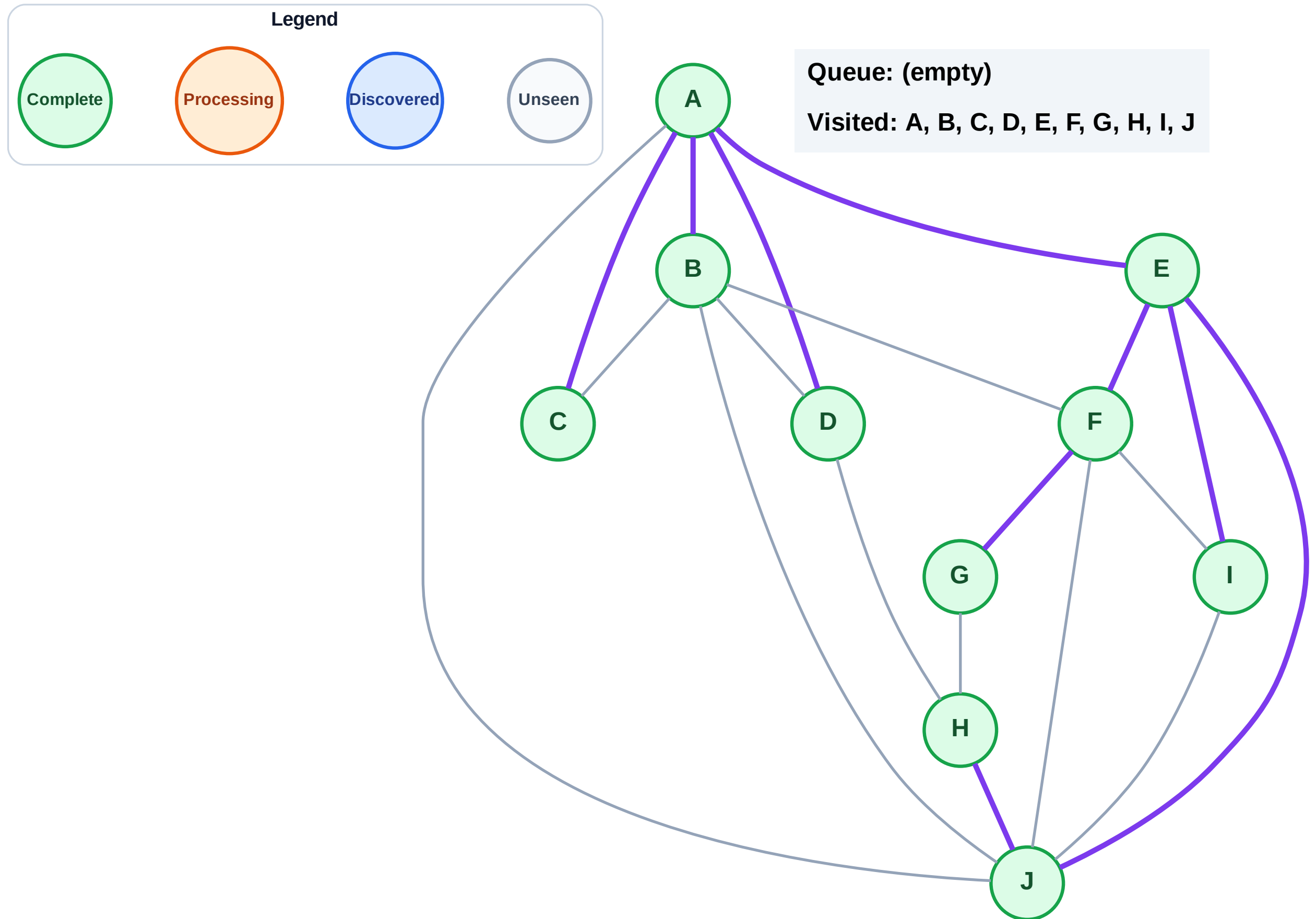
Queue: (empty)

Visited: A, B, C, D, E, F, G, I, J



Step 21: No new neighbors from H

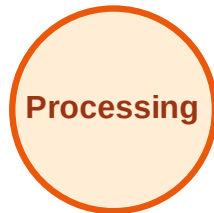
Step 21: No new neighbors from H



DFS Traversal

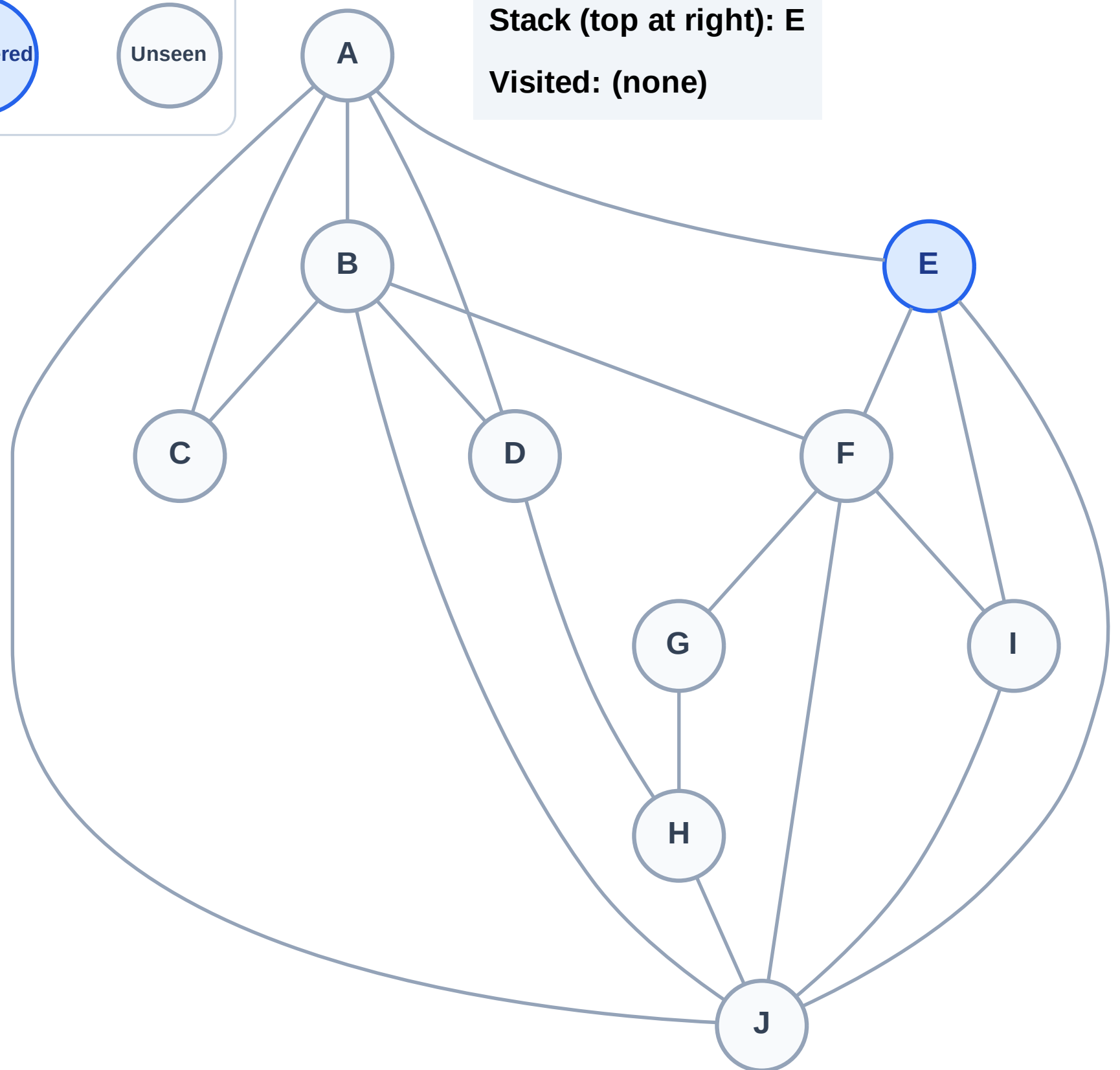
Step 1: Start at E

Legend



Stack (top at right): E

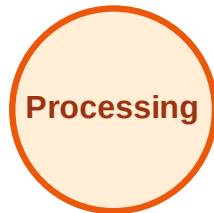
Visited: (none)



DFS Traversal

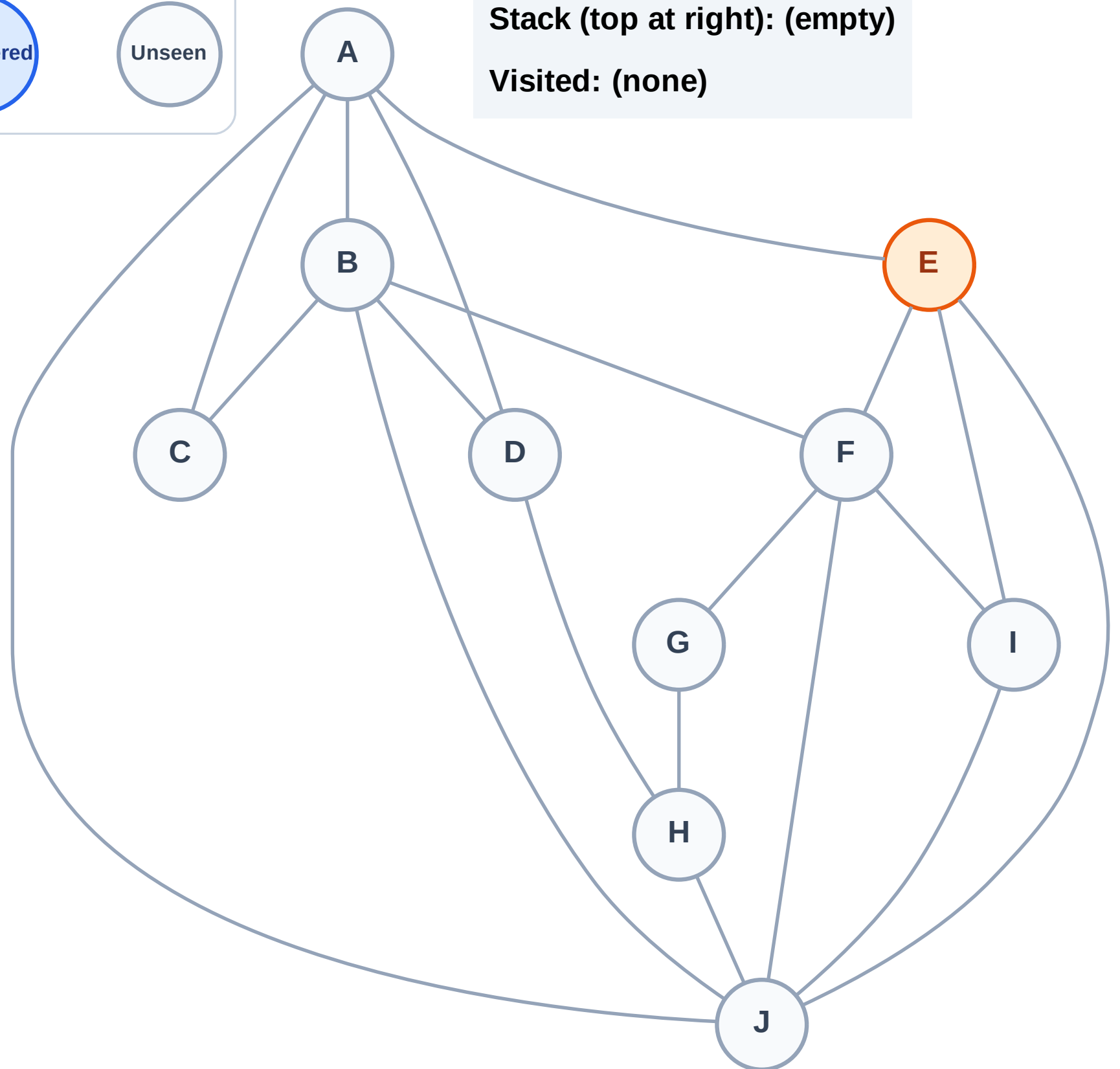
Step 2: Process node E

Legend



Stack (top at right): (empty)

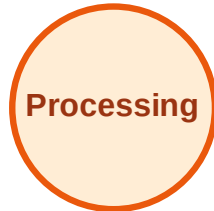
Visited: (none)



DFS Traversal

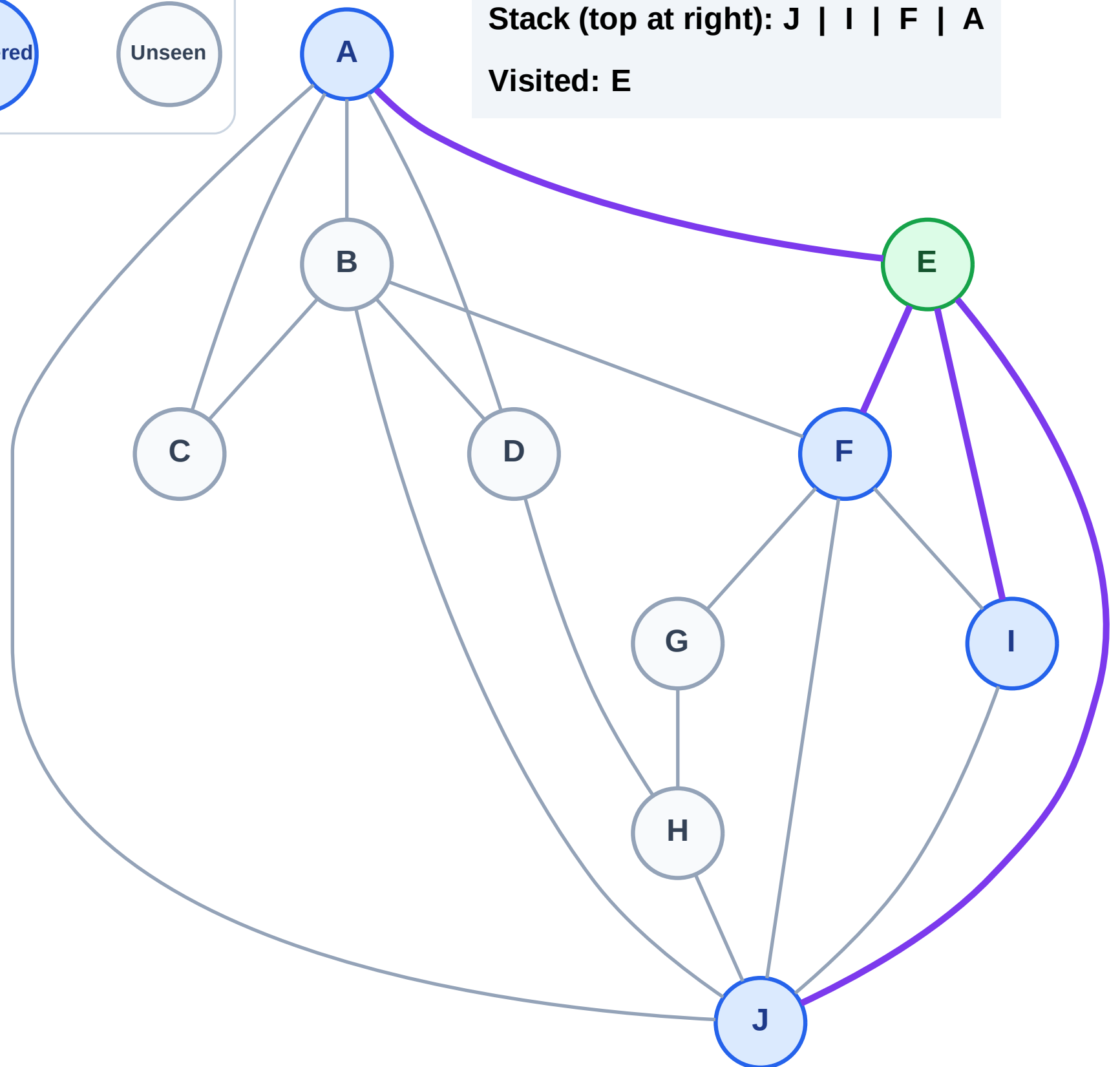
Step 3: Discover J, I, F, A from E

Legend



Stack (top at right): J | I | F | A

Visited: E



DFS Traversal

Step 4: Process node A

Legend

Complete

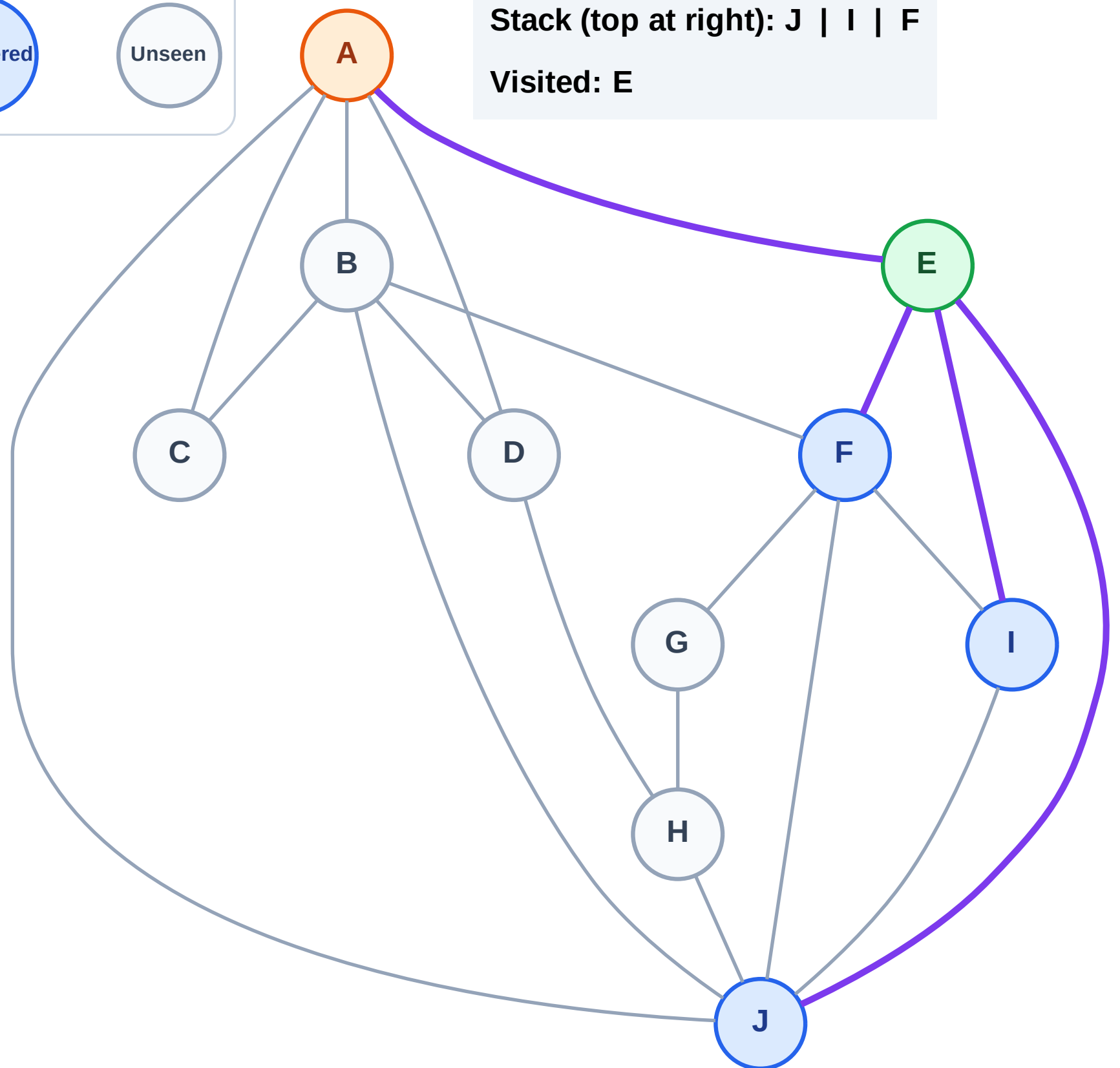
Processing

Discovered

Unseen

Stack (top at right): J | I | F

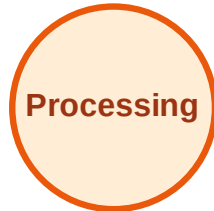
Visited: E



DFS Traversal

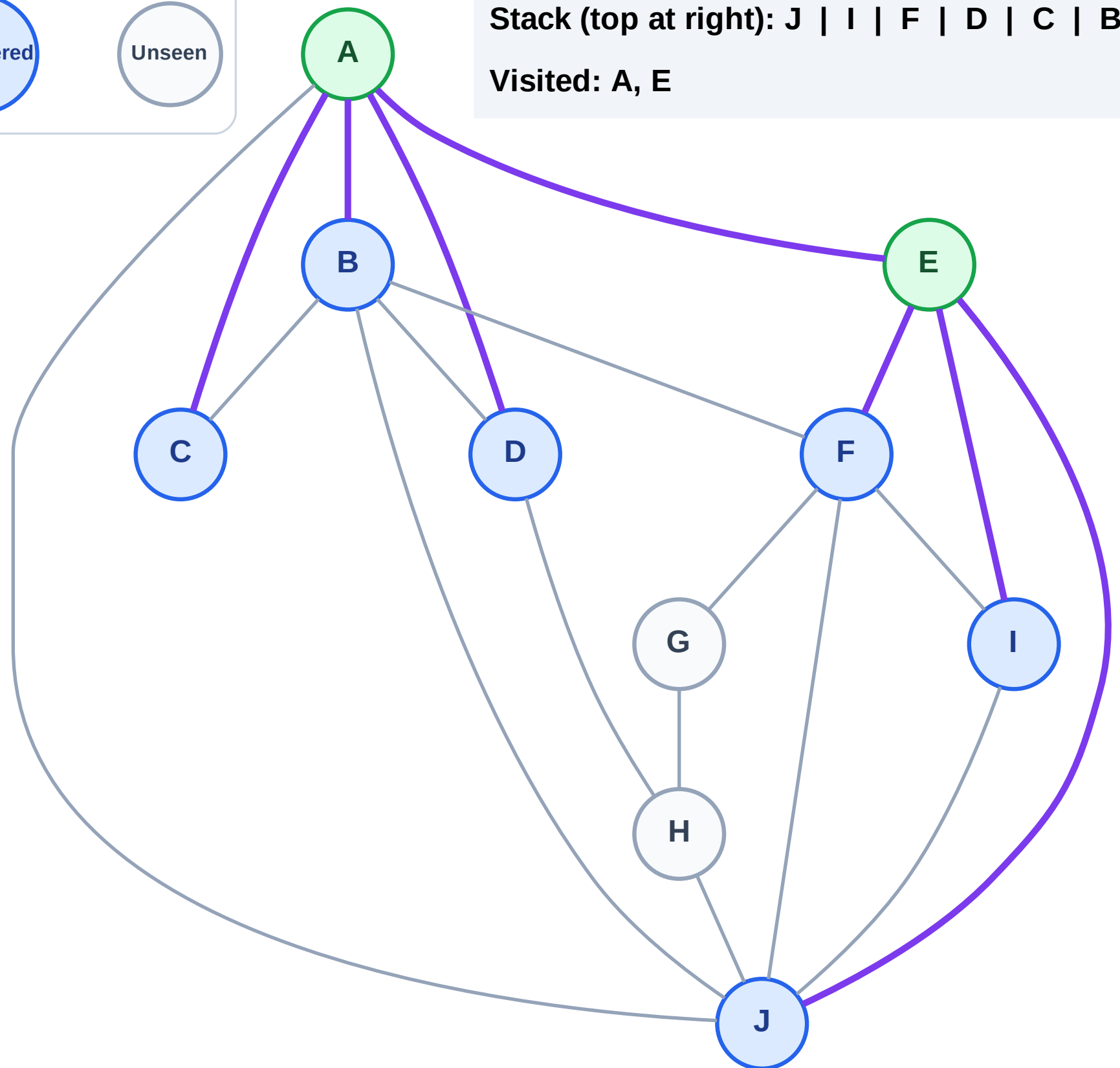
Step 5: Discover D, C, B from A

Legend



Stack (top at right): J | I | F | D | C | B

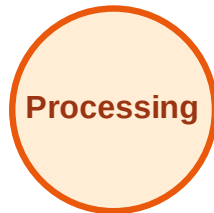
Visited: A, E



DFS Traversal

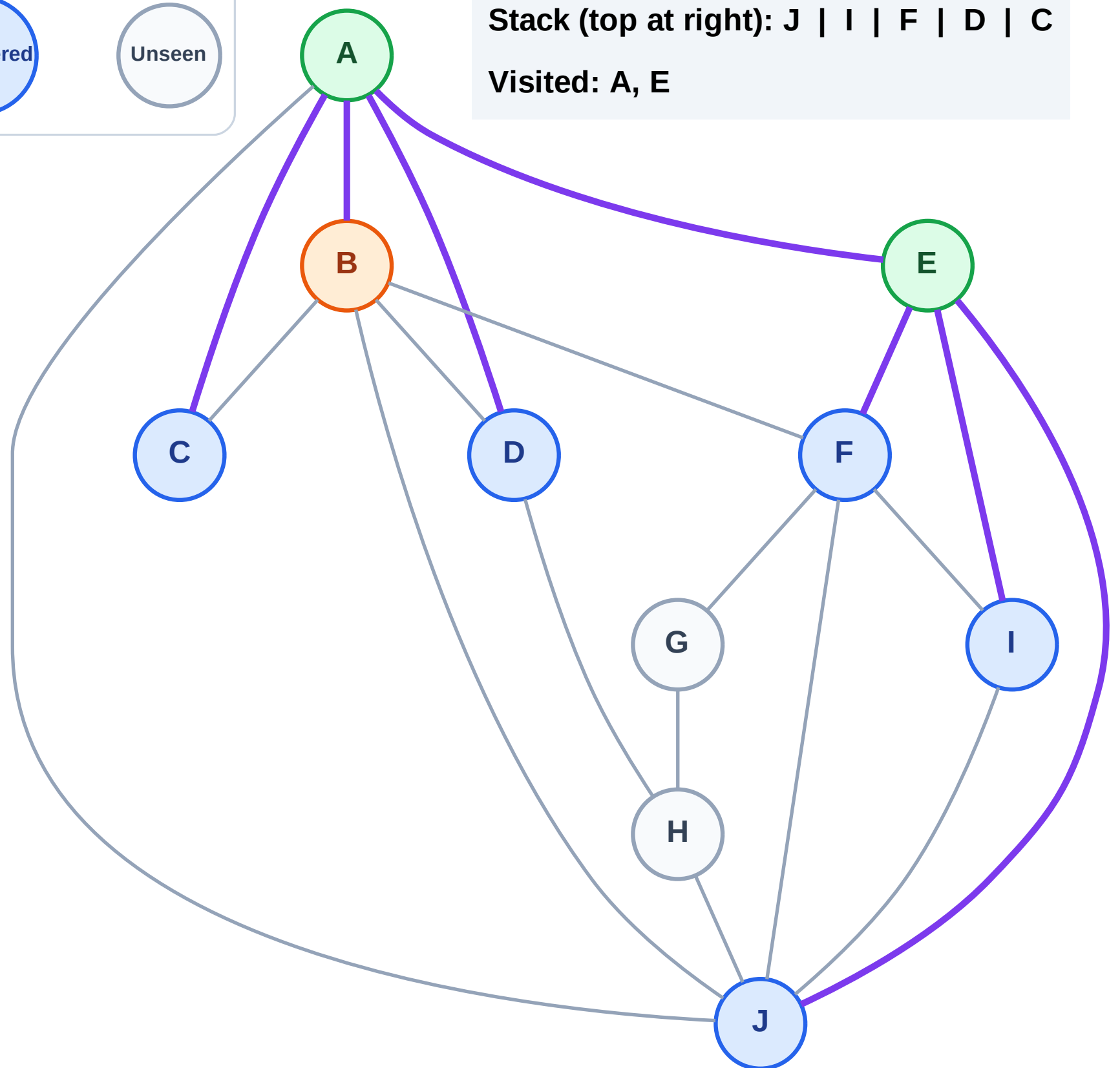
Step 6: Process node B

Legend



Stack (top at right): J | I | F | D | C

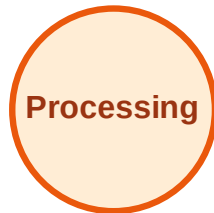
Visited: A, E



DFS Traversal

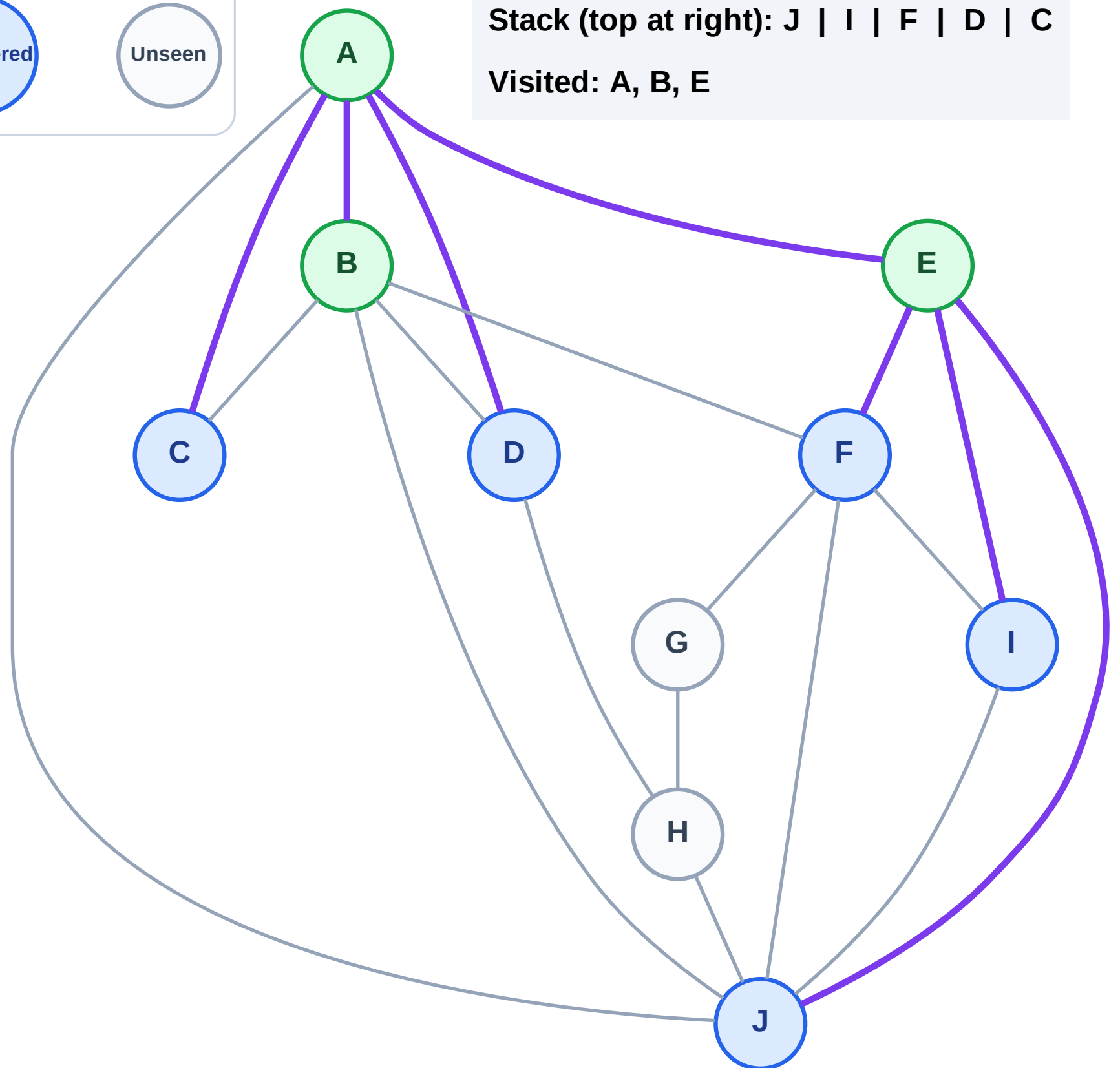
Step 7: No new neighbors from B

Legend



Stack (top at right): J | I | F | D | C

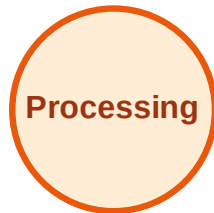
Visited: A, B, E



DFS Traversal

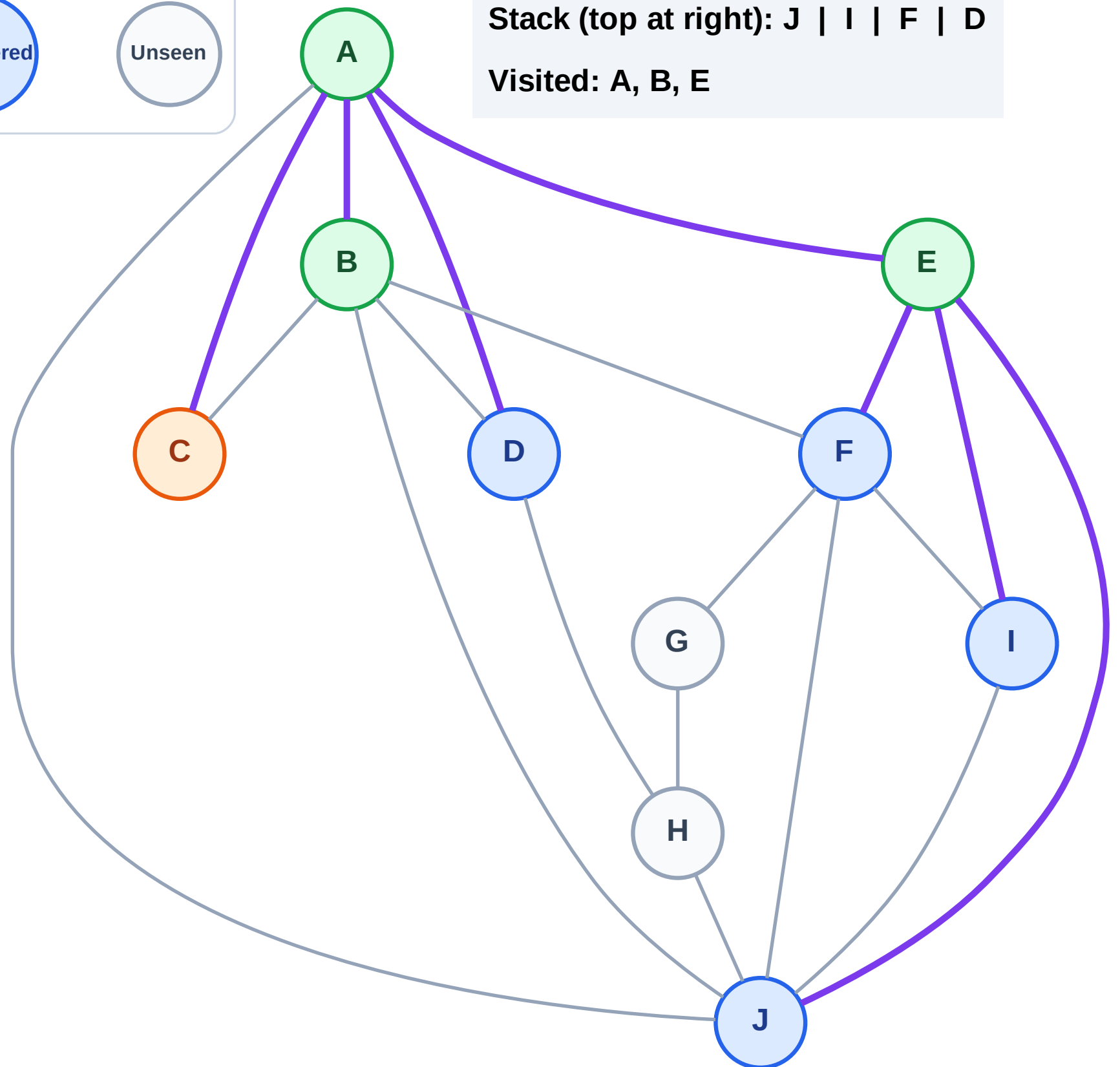
Step 8: Process node C

Legend



Stack (top at right): J | I | F | D

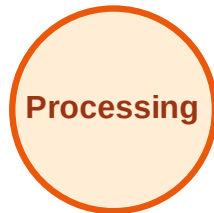
Visited: A, B, E



DFS Traversal

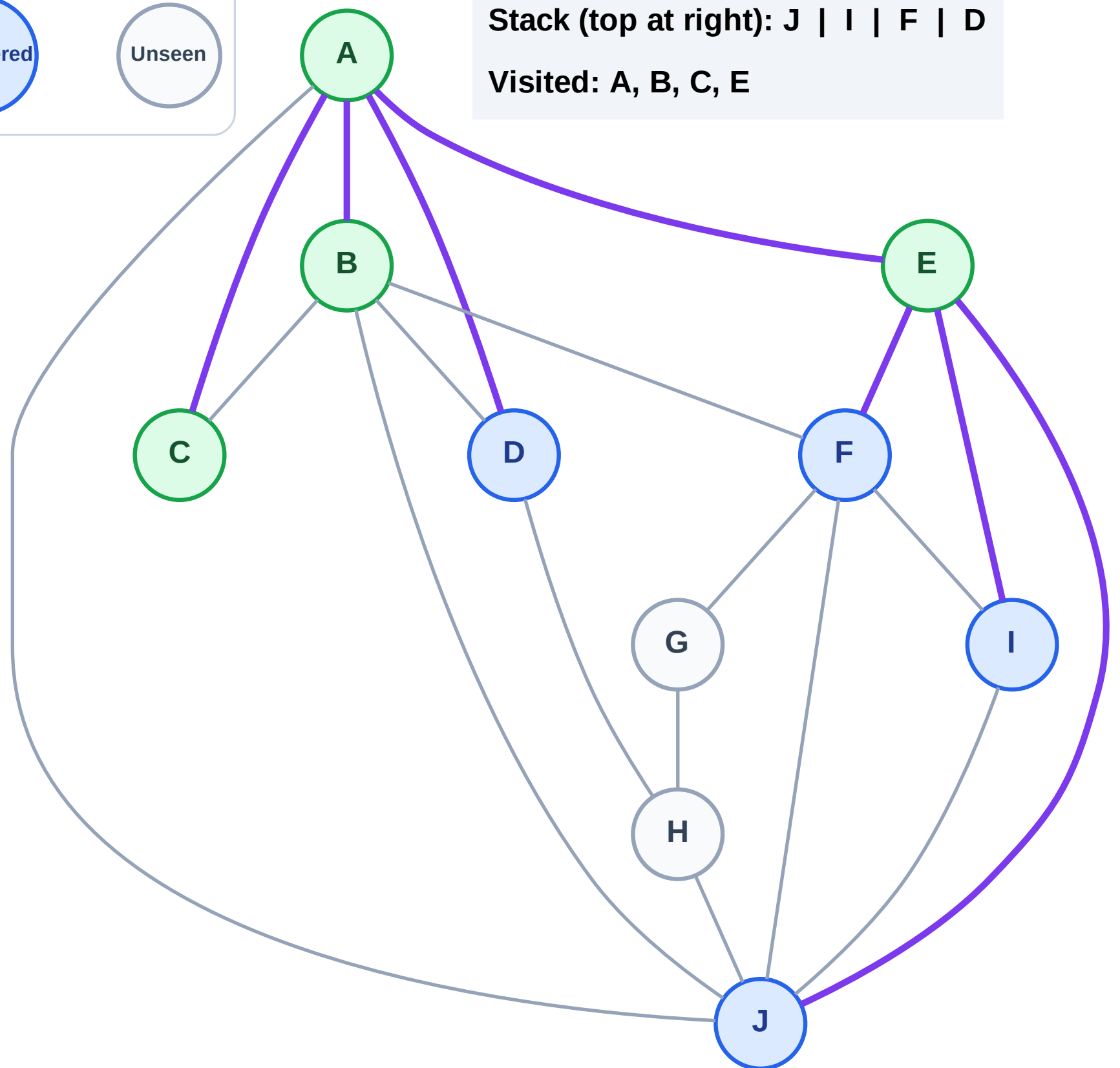
Step 9: No new neighbors from C

Legend



Stack (top at right): J | I | F | D

Visited: A, B, C, E



DFS Traversal

Step 10: Process node D

Legend

Complete

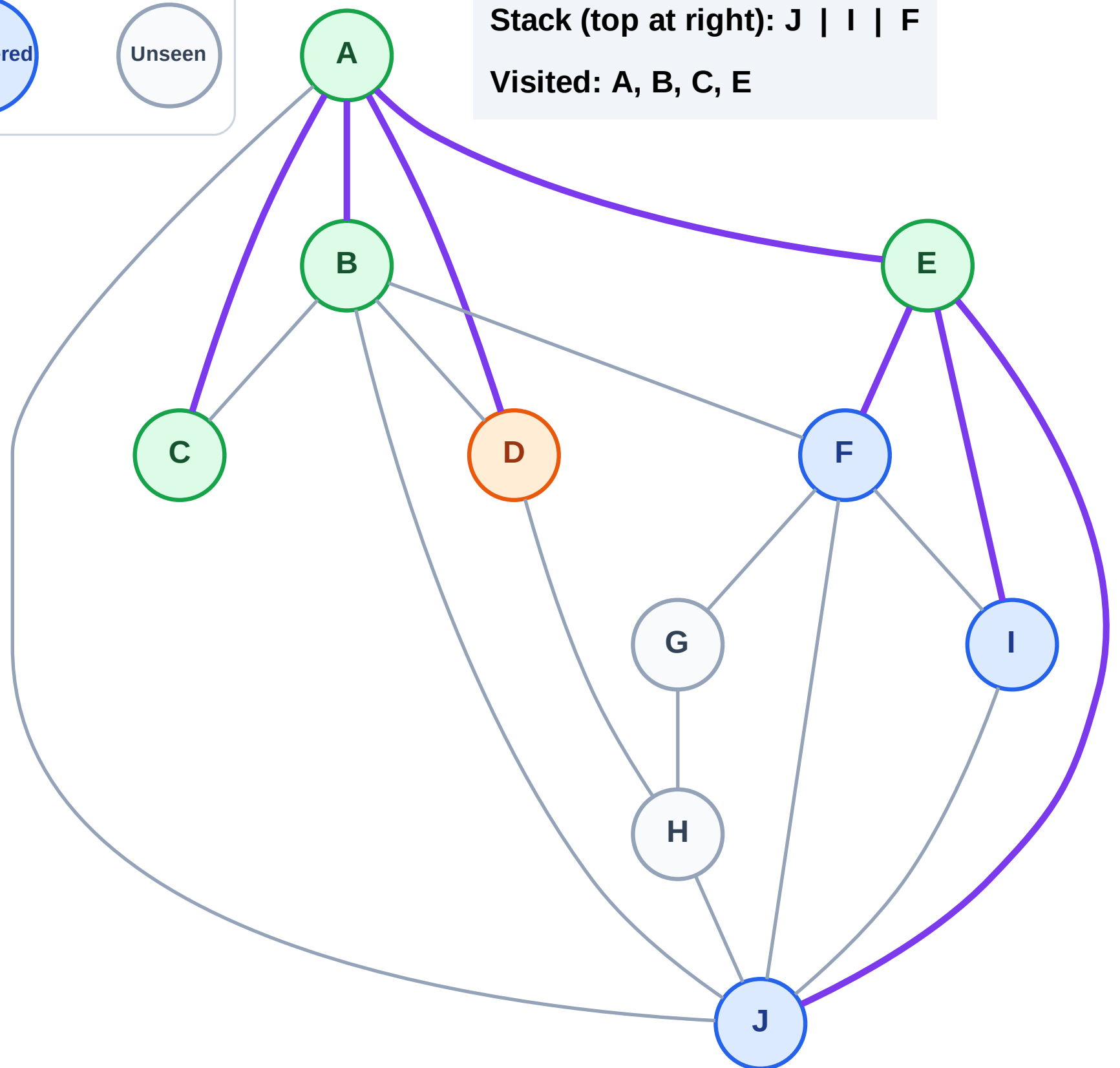
Processing

Discovered

Unseen

Stack (top at right): J | I | F

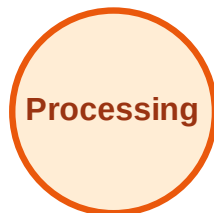
Visited: A, B, C, E



DFS Traversal

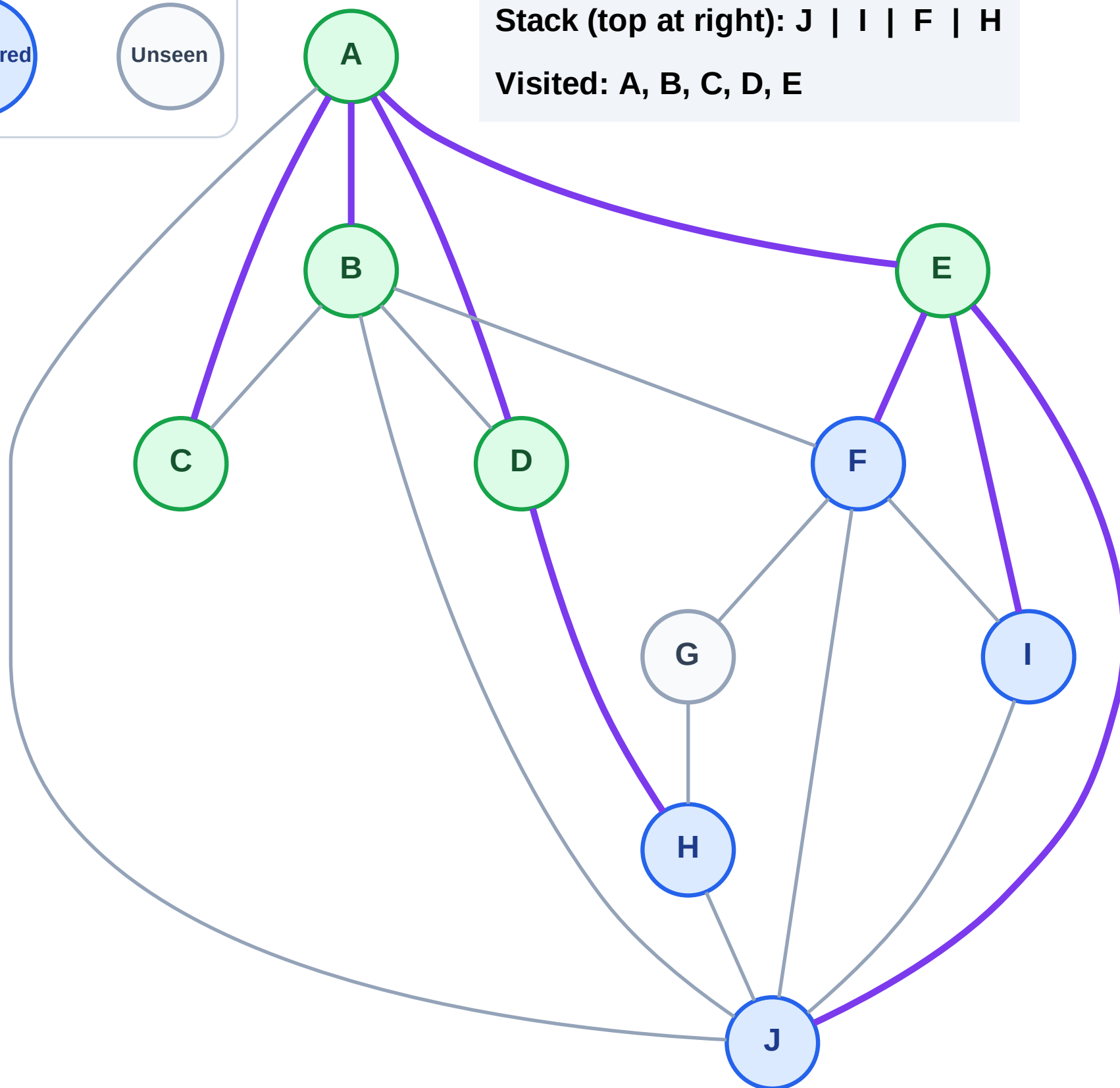
Step 11: Discover H from D

Legend



Stack (top at right): J | I | F | H

Visited: A, B, C, D, E



DFS Traversal

Step 12: Process node H

Legend

Complete

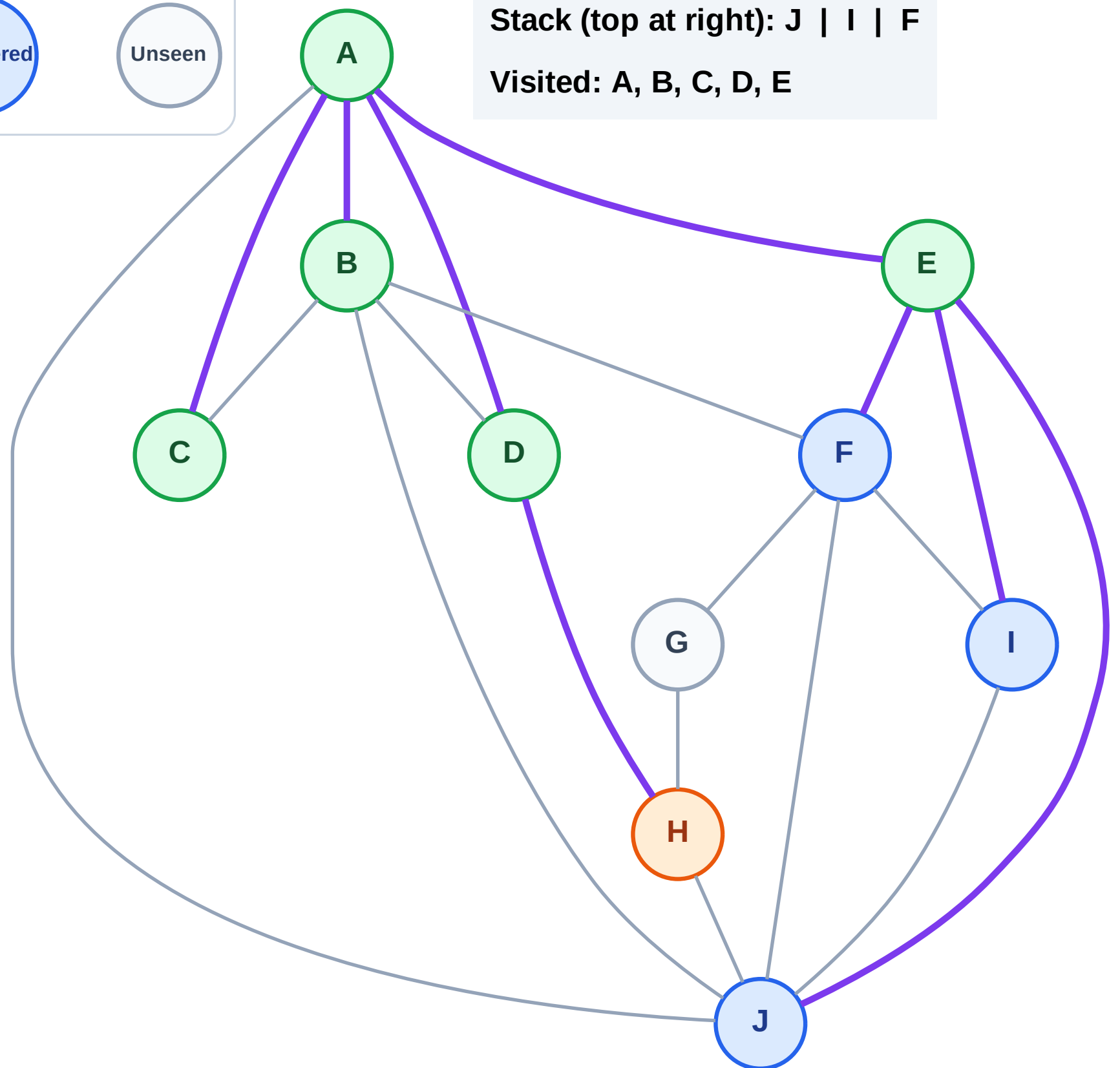
Processing

Discovered

Unseen

Stack (top at right): J | I | F

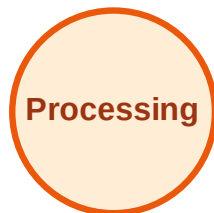
Visited: A, B, C, D, E



DFS Traversal

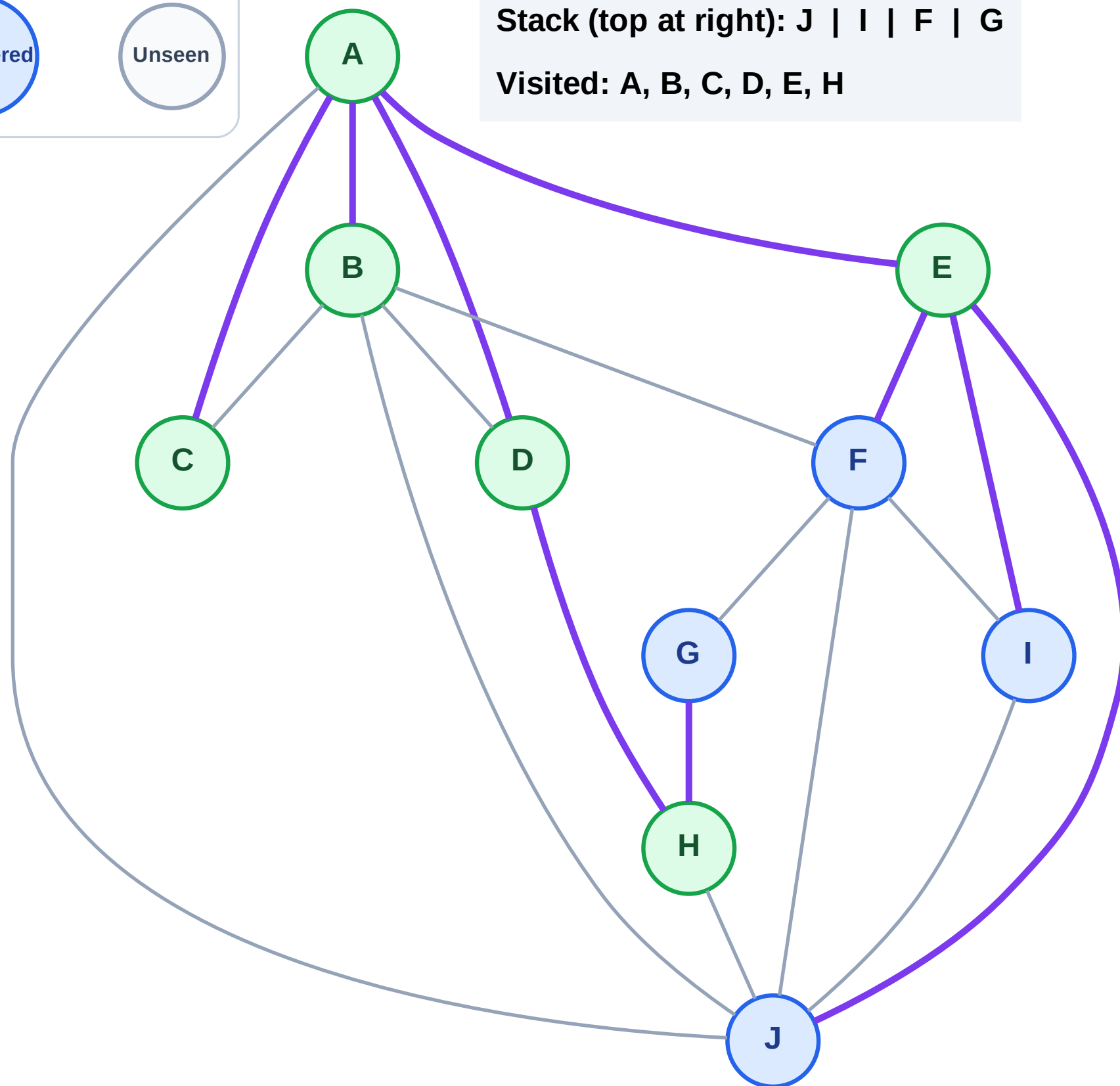
Step 13: Discover G from H

Legend



Stack (top at right): J | I | F | G

Visited: A, B, C, D, E, H



DFS Traversal

Step 14: Process node G

Legend

Complete

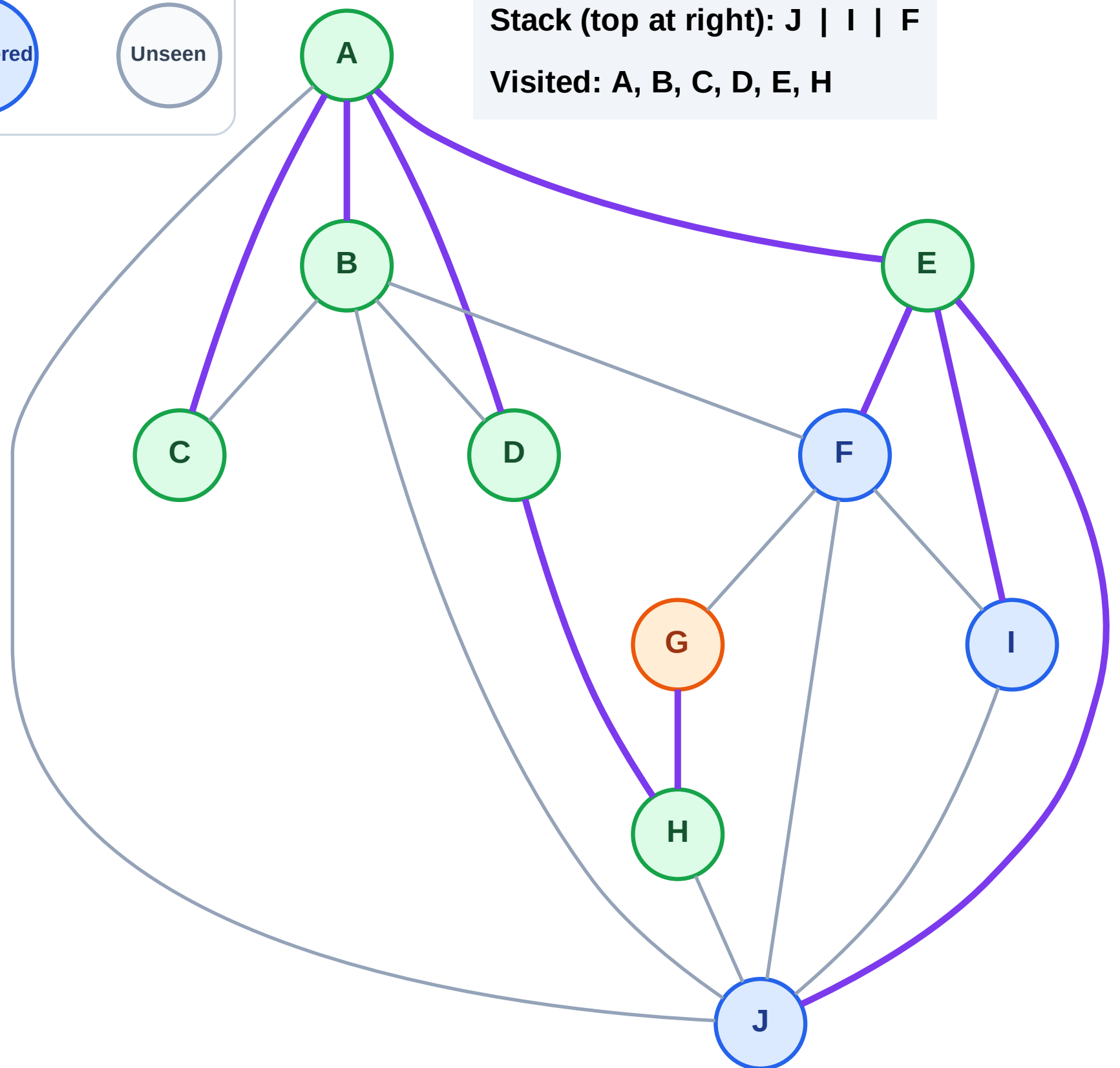
Processing

Discovered

Unseen

Stack (top at right): J | I | F

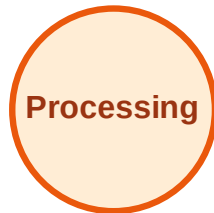
Visited: A, B, C, D, E, H



DFS Traversal

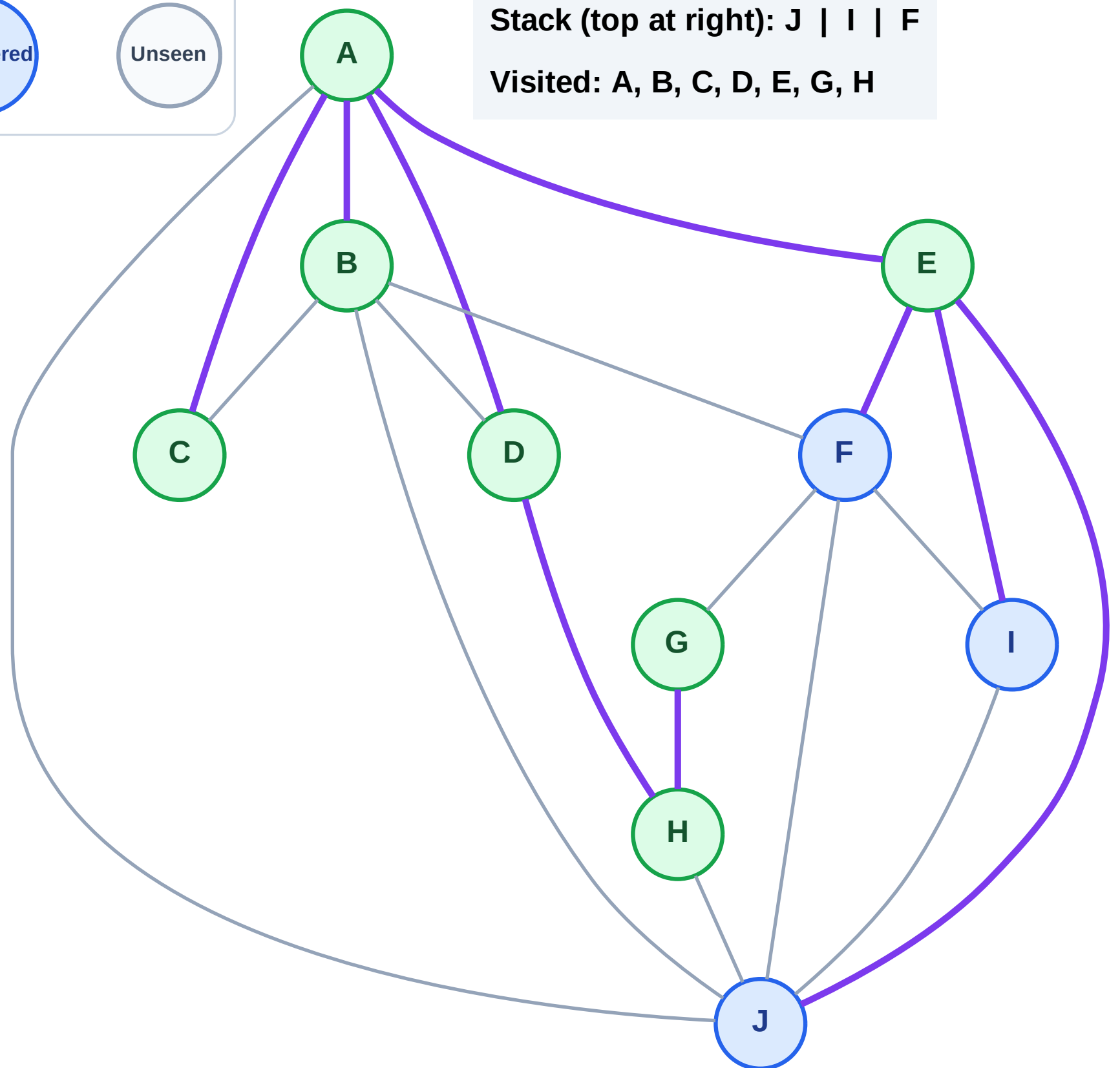
Step 15: No new neighbors from G

Legend



Stack (top at right): J | I | F

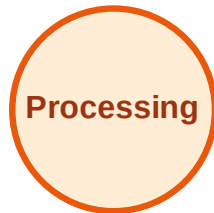
Visited: A, B, C, D, E, G, H



DFS Traversal

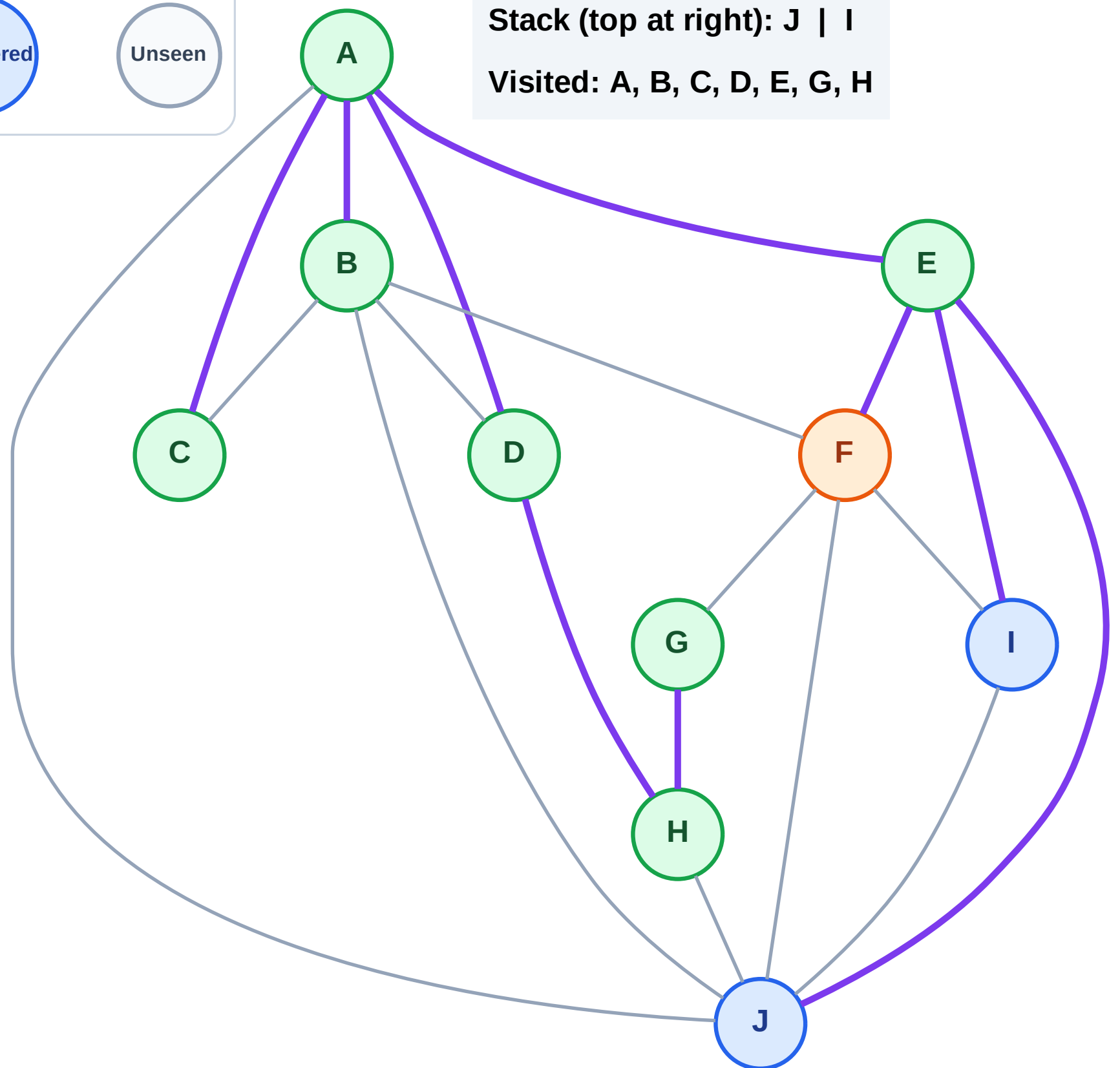
Step 16: Process node F

Legend



Stack (top at right): J | I

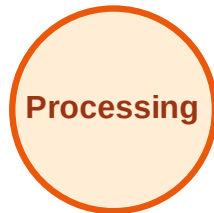
Visited: A, B, C, D, E, G, H



DFS Traversal

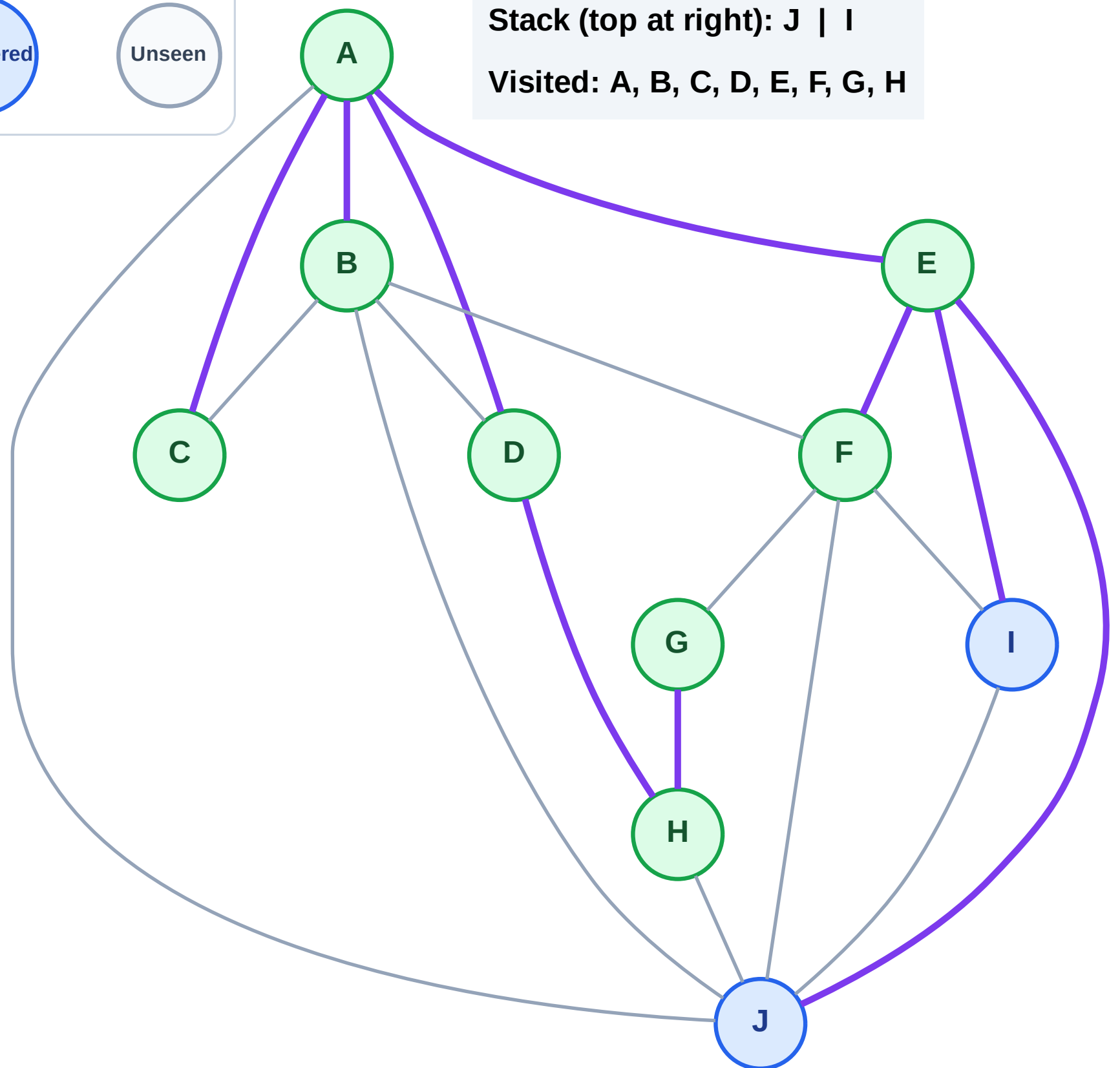
Step 17: No new neighbors from F

Legend



Stack (top at right): J | I

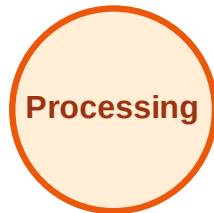
Visited: A, B, C, D, E, F, G, H



DFS Traversal

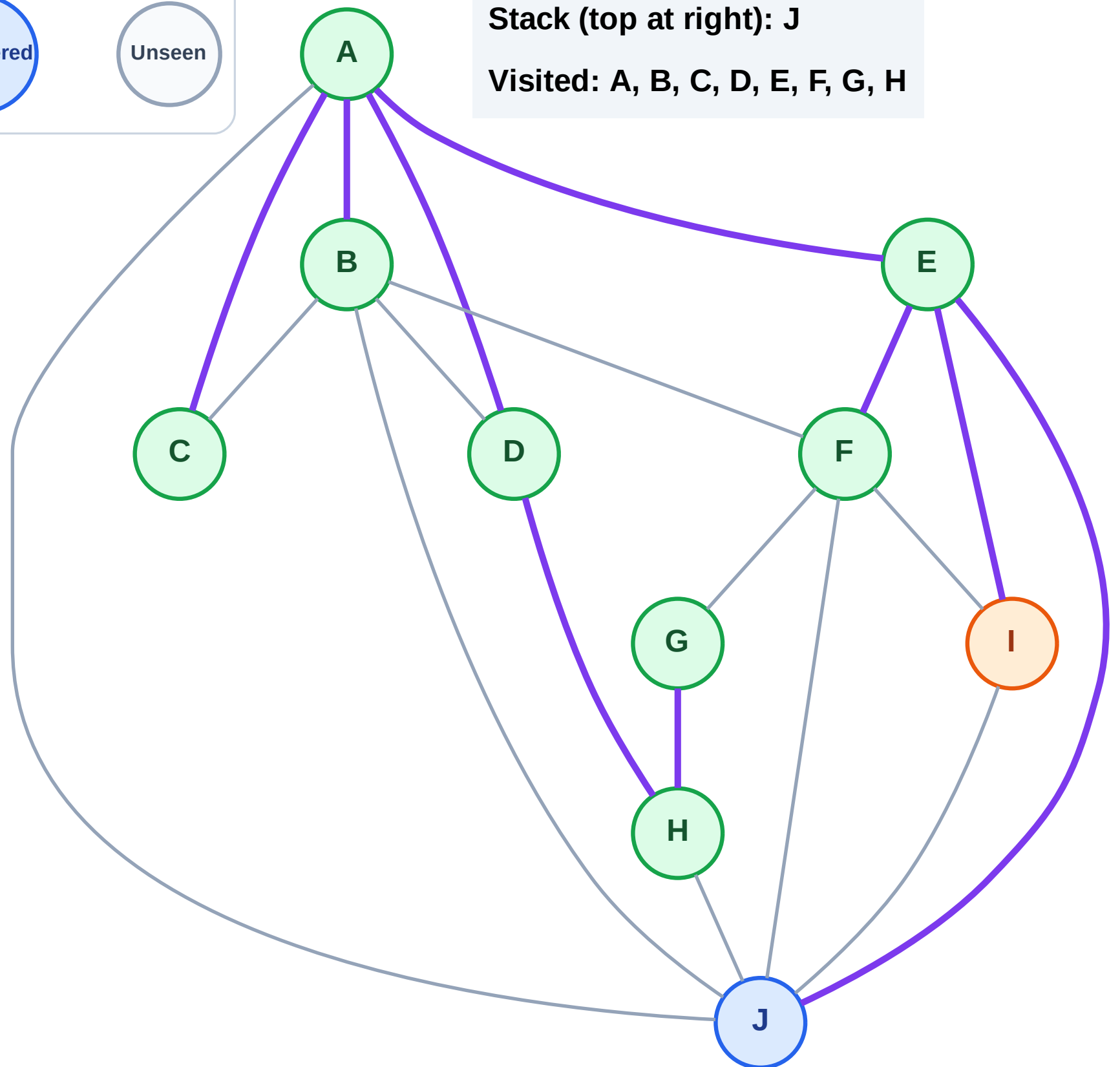
Step 18: Process node I

Legend



Stack (top at right): J

Visited: A, B, C, D, E, F, G, H



DFS Traversal

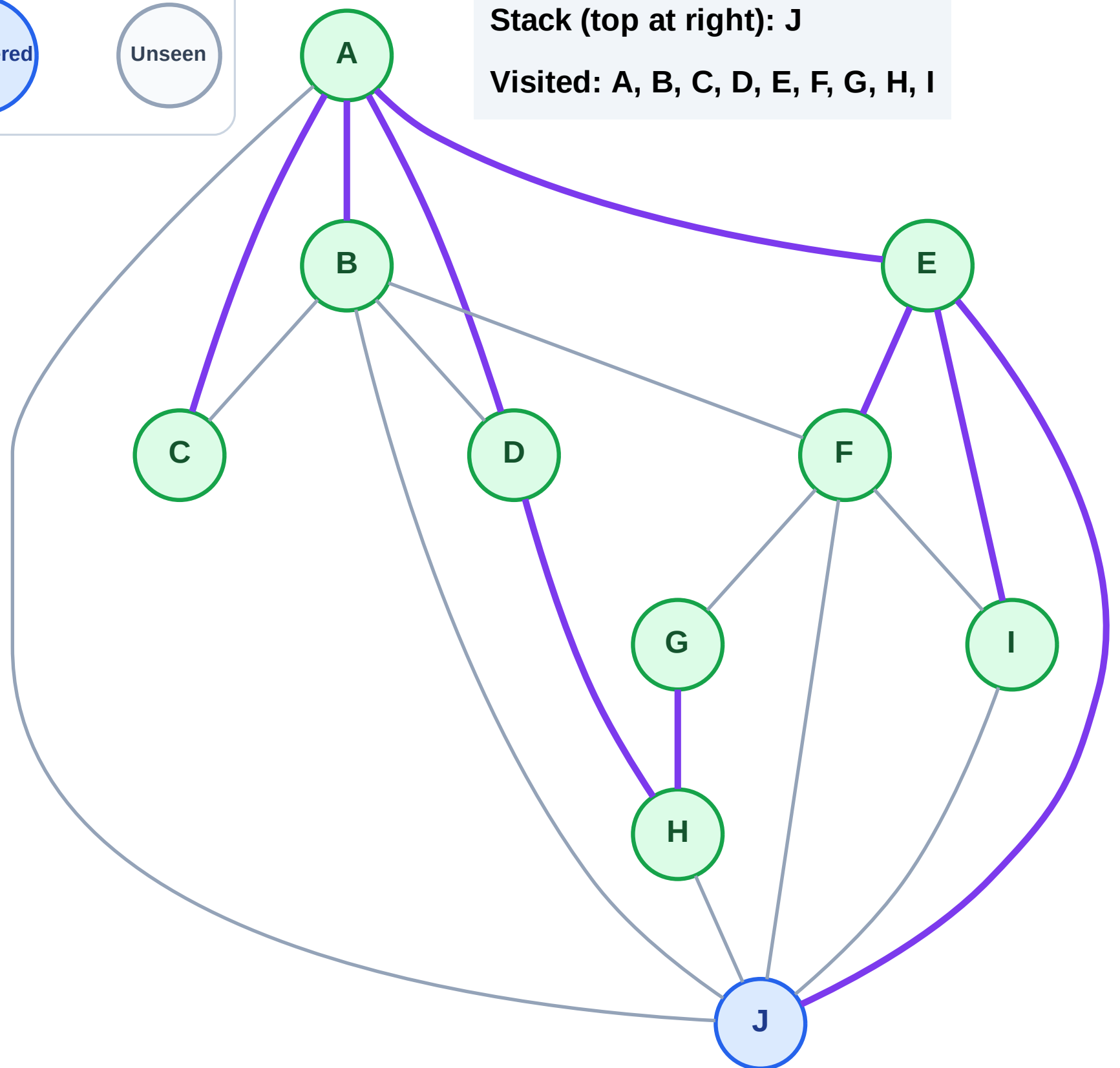
Step 19: No new neighbors from I

Legend



Stack (top at right): J

Visited: A, B, C, D, E, F, G, H, I



DFS Traversal

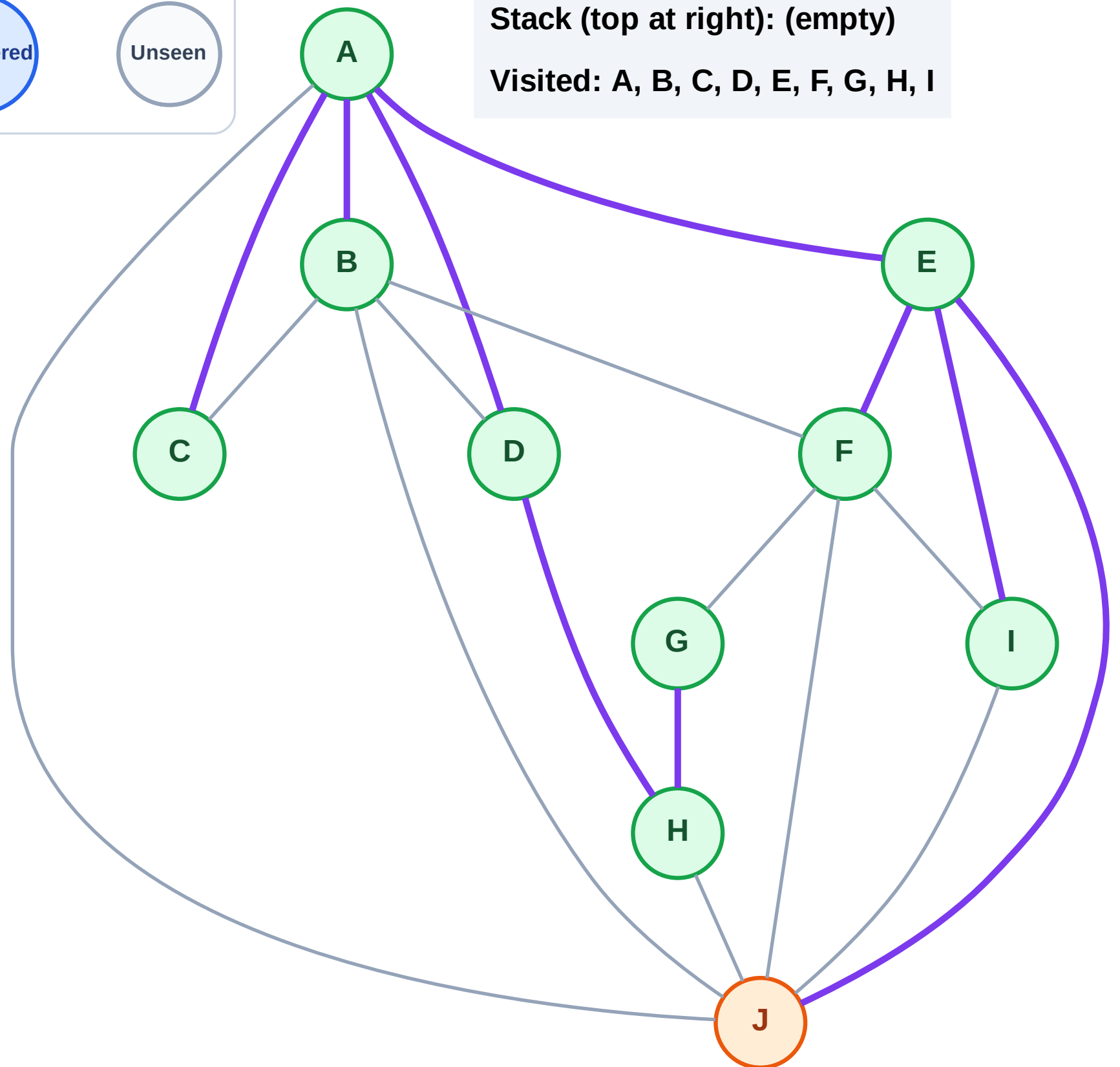
Step 20: Process node J

Legend



Stack (top at right): (empty)

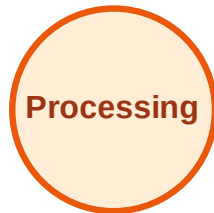
Visited: A, B, C, D, E, F, G, H, I



DFS Traversal

Step 21: No new neighbors from J

Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I, J

